

Insights from the
Global Education
Policy Dashboard

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KEY TAKEAWAYS

- 1 Foundational skills are the bedrock of the skills-for-jobs agenda.** Without strong literacy, numeracy, and socio-emotional skills acquired in primary school, young people cannot build the job-relevant skills that economies need. Yet across low- and middle-income countries, 70 percent of children do not master the basics by the end of primary school.¹ Delivering on the promise of “skills for jobs” starts with fixing what happens in the early grades.
- 2 Improving foundational skills starts with an understanding of the education systems that deliver them.** Yet many countries lack the integrated diagnostic tools required. The Global Education Policy Dashboard (GEPD) was designed to meet this need, measuring school-level practices, policy implementation, and political and bureaucratic conditions through a single integrated tool. The approach is validated by the data: together, these three layers explain about 73 percent of the variation in Grade 4 student achievement across participating systems. Furthermore, the approach is cost-effective, averaging at less than half the cost of earlier instruments.
- 3 GEPD data reveal that progress is achievable, despite significant gaps in the school-level practices and policies needed to deliver foundational skills.** Across most systems surveyed, fewer than 20 percent of teachers demonstrate content proficiency, and many children arrive at school without basic readiness skills. Yet the data also show that meaningful gains are achievable: in Ethiopia, teachers absence from the school premises halved and school readiness rose from 33 to 45 percent in just five years.
- 4 Closing these gaps depends on education officials who design policies, allocate resources, and manage personnel—and most are committed public servants.** Eighty-one percent of public officials report satisfaction with public service. Yet structural weaknesses undermine their efforts: one in four report that hiring favors political connections over merit, and about half say their education systems lack the data and incentives needed to track and reward progress on foundational learning.
- 5 Acting on these insights requires first knowing where the problem lies, and that requires data.** In Chad and Sierra Leone, most variation in learning outcomes is between schools, pointing to a need to raise the floor for the lowest-performing schools. In Montenegro and Pakistan’s Islamabad Capital Territory, the reverse is true: most variation is within the same classroom, calling for targeted instruction and differentiated support. Without system-level diagnostics, these distinctions remain invisible.
- 6 Foundational skills are essential, but they alone cannot prepare workers for the jobs of tomorrow.** To build the full skills pipeline economies need, countries need to extend a similar diagnostic approach used by the GEPD to secondary, technical and vocational education and training (TVET), and higher education. The GEPD’s system-oriented framework offers a strong basis for future diagnostic work to understand and strengthen skills development beyond primary grades.

EXECUTIVE SUMMARY

Addressing the Learning Crisis Requires Regular and Reliable Data on Education Systems

Education and skills drive economic growth and job creation. The stakes are high: over the next decade, an estimated 1.2 billion young people will enter the workforce in emerging economies. Whether they find productive employment, and whether countries can meet their development ambitions in key sectors, such as infrastructure and energy, agribusiness, healthcare, tourism, and value-added manufacturing, will depend on how well education systems equip learners with the foundational and job-relevant skills required in a rapidly changing labor market.

Yet for most children, years of schooling yield little actual learning. Despite substantial progress in expanding access to school, learning outcomes remain alarmingly low. Roughly 70 percent of children in low- and middle-income countries are unable to read and understand a basic text by the end of primary school, based on the latest available simulated Learning Poverty estimates.² Without strong foundational literacy, numeracy, and socio-emotional skills, young people are unlikely to acquire job-relevant skills, and enter the workforce with the skills needed to be productive, resilient, and adaptable. This undermines both individual opportunity and broader economic development.

Understanding why learning outcomes remain low requires looking beyond individual interventions to the systems in which they operate. Education systems are complex: policies designed at the center must cascade through multiple administrative levels, be interpreted by officials with varying capacities, and ultimately shape daily decisions in diverse classroom contexts. Even well-designed reforms can fail when different parts of the system are not well aligned. For example, a curricular reform will not improve student learning when teachers do not receive relevant training; when textbooks, lesson plans, and assessments are misaligned with curricular goals; when instructional support is weak; or when officials lack data and mechanisms to monitor implementation and progress. Addressing the learning crisis therefore requires not only effective interventions, but also better visibility into the system-level factors that determine whether those interventions succeed.

Yet, many education systems, especially in low- and middle-income countries, lack a single, integrated diagnostic tool capable of systematically examining the drivers of learning across school-level practices, policies, and political and bureaucratic conditions. While system-level data are essential for effective reform, such data are often unavailable or fragmented, offering limited insight into what is working well and what is not.

The Global Education Policy Dashboard (GEPD) Provides Actionable Data to Improve Education Systems

The World Bank launched the GEPD in 2019 to provide system-level visibility on the drivers of learning in primary school. Building on earlier instruments such as the Service Delivery Indicators (SDI) and the Systems Approach for Better Education Results (SABER), the GEPD offers a unified, system-oriented framework to understand opportunities and challenges to improve learning. To date, the GEPD has been implemented in 27 education systems, representing around 18 percent of World Bank IDA/Blend countries.ⁱ Additional implementations are underway.ⁱⁱ This report draws on analysis of available school survey data from 18 education systems,ⁱⁱⁱ and public officials' data from 19 education systems. Participating systems span a wide range of income levels and regions, including Latin America, Sub-Saharan Africa, South Asia, the Middle East, and Europe. In some systems, GEPD is the only integrated source of information on learning and its drivers. For example, in eight of the systems surveyed, there is no recent learning assessment data available to compute Learning Poverty estimates.

This report analyzes data from GEPD implementations to date to take stock of progress and challenges in improving learning. It answers three core questions:

- 1. What does the GEPD measure?** The report establishes the case for investing in foundational skills, which are the bedrock of the skills-for-jobs agenda, and argues that delivering them requires system-level data on the drivers of learning. It also describes the GEPD tools, including the underlying conceptual framework and instruments.
- 2. What do data from GEPD show regarding learning outcomes and their underlying drivers across countries?** The report analyzes data from participating systems and provides a synthesis of cross-country evidence on learning outcomes and their key drivers, along with insights from interviews with public officials.
- 3. How can GEPD data be used to support diagnosis, policy dialogue, and reform implementation?** The report discusses the use and limitations of the data, how governments and World Bank teams can use the data to inform future work, and closes with key messages for the way forward.

The findings are intended for a broad audience, including policy makers, development partners, and World Bank teams engaged in education and skills development. By bringing together evidence on classroom and school practices, policy design and implementation, and political and bureaucratic conditions, the findings from the GEPD provide a holistic view of how education systems function in practice. The findings not only show where learning outcomes are weak, but also where gaps in school-level practices, policy design and implementation, and leadership and bureaucratic capacity constrain progress.

ⁱIDA-eligible countries are the world's poorest, receiving grants and low-interest loans (credits). "Blend" countries are a sub-category that are IDA-eligible but creditworthy enough to also borrow from IBRD (International Bank for Reconstruction and Development).

ⁱⁱ27 completed implementations (in chronological order of implementations): Peru (1st round), Rwanda, Jordan (1st round), Ethiopia (1st round), Madagascar, Mozambique (partial), Edo State (Nigeria), Sindh (Pakistan), Punjab (Pakistan), KP (Pakistan), Islamabad CT (Pakistan), Balochistan (Pakistan), Niger, Gabon, Chad, Sierra Leone, Jordan (2nd round), Bangladesh, the Central African Republic, Papua New Guinea, Ethiopia (2nd round), Federal Republic of Somalia, Montenegro, Colombia, Peru (2nd round), Seychelles, and Togo.

ⁱⁱⁱThe number of systems covered may be lower for some variables, depending on whether data was collected on those aspects. For example, for Ethiopia, no Policy Survey was completed in the 2025 round. For Madagascar, no TEACH Primary data is available. For Mozambique, only data from the public officials survey is available.

GEPD Data Reveal Persistent Gaps Across Key Drivers of Foundational Learning

Drawing on the framework advanced in the World Development Report (2018), this report examines the drivers of learning across three levels:

1. **School-level practices**, including learner preparation, teacher content knowledge and pedagogy, availability of school inputs and infrastructure, and school management;
2. **Policy levers**, including gaps between *de jure* policies and their *de facto* implementation; and
3. **Politics and bureaucratic capacity**, including leadership turnover, incentives, information systems, and decision-makers' knowledge of frontline realities.

Across the majority of systems surveyed, fewer than 10 percent of children have mastered the literacy and numeracy skills expected by Grade 4, ranging from almost no children achieving proficiency in the Central African Republic, Chad, Niger, Rwanda to more than 30 percent in Montenegro and Peru. This finding is consistent with the high rates of Learning Poverty documented across many of the surveyed systems, including Chad, Ethiopia, Madagascar, and Niger, where more than 90 percent of children are unable to read with comprehension by the end of primary school.^{iv}

In some systems, school quality varies widely, while in others, most of the differences in learning outcomes are inside the same classroom. For example, in Chad and Sierra Leone, most of the variation in learning outcomes is between schools. In Montenegro and Pakistan's Islamabad Capital Territory (ICT), the pattern reverses.^v These patterns have policy implications. Systems where between-school variation is high should prioritize raising performance in the lowest-performing schools. Meanwhile, systems where within-school variation dominates should focus on targeted instruction and differentiated support to meet students where they are.

The findings highlight persistent gaps in school-level practices, policy implementation, and leadership and bureaucratic capacity that shape learning outcomes. They are summarized below.

School-Level Practices

- **LEARNER PREPARATION.** Children's readiness when they enter school is one of the strongest predictors of later learning outcomes. Yet many children arrive at school without the skills needed to thrive during primary education. Only in three systems surveyed (Bangladesh, Montenegro, and Peru) do more than 50 percent of the students demonstrate mastery of basic skills, such as emerging literacy and numeracy, executive functioning, and socio-emotional skills, though this varies considerably across contexts. In Montenegro, more than 70 percent of children enter Grade 1 with mastery of basic skills, while other systems, such as the Central African Republic, Madagascar, and Pakistan's Islamabad Capital Territory (ICT), Khyber Pakhtunkhwa (KP), and Sindh face steeper challenges, with readiness rates in single digits. However, even in challenging contexts, progress is achievable. In Ethiopia, school readiness improved from 33 percent to 45 percent between 2020 and 2025, demonstrating that meaningful gains are possible within a few years with the right investments.

^{iv} Learning Poverty (LP) and Learning Deprivation (LD) rates: Bangladesh (LP: 51 percent; LD: 50 percent), Chad (LP: 94 percent; LD: 92.4 percent), Ethiopia (LP: 90 percent; LD: 88.7 percent), Gabon (LP: 31 percent; LD: 23.7 percent), Jordan (LP: 62 percent; LD: 53.4 percent), Madagascar (LP: 95 percent; LD: 93.7 percent), Niger (LP: 90 percent; LD: 85.6 percent), Pakistan (LP: 80 percent; LD: 70 percent), Peru (LP: 44 percent; LD: 43.6 percent). Retrieved November 3, 2025 from <https://www.worldbank.org/en/topic/education/brief/country-learning-poverty-briefs>.

^v Throughout this report ICT means Islamabad Capital Territory.

- **TEACHER EFFECTIVENESS.** Effective student learning relies on teachers who are present in the classroom and have strong content knowledge and pedagogical skills. While the data reveal critical gaps, they also show some signs of progress. Teacher presence has risen across some systems, showing that progress is possible. For example, data from Ethiopia show that the proportion of teachers absent from the school premises halved between 2020 and 2025, falling from 17 percent to 9 percent, though classroom presence proved harder to improve. Overall, across most systems surveyed, less than 20 percent of teachers demonstrate proficiency in the subjects they teach, ranging from under 10 percent across seven systems to 60 percent (Bangladesh). Furthermore, across most education systems, at least half of the variation in teacher content knowledge occurs among teachers within the same school, underscoring the need for tailored school-based coaching, mentoring, and peer-based learning resources. As is the case with teacher content knowledge, the majority of teachers do not have satisfactory pedagogical skills across most education systems, with Gabon, Jordan, and Pakistan's ICT showing relatively better performance.
- **INPUTS AND INFRASTRUCTURE.** Effective learning requires schools to be equipped with essential resources. Yet wide disparities persist in access to essential infrastructure for education service delivery. Over the past few decades, many countries have invested heavily in school infrastructure as part of broader efforts to expand access to education. This progress is visible in the data. For instance, in Ethiopia, the urban-rural electrification gap decreased by approximately 24 percentage points between 2020 and 2025. Ethiopia is not an isolated example, and some countries have even attained near-universal coverage. Meanwhile, in the Central African Republic, Chad, Niger, and Sierra Leone, by contrast, coverage remains at under 10 percent of schools. The picture for basic learning inputs is uneven. Inputs such as textbooks, pens, and exercise books are more than classroom essentials. They are tools that enable students to engage with the content being taught. Yet in the majority of the systems surveyed, less than 80 percent of Grade 4 students have textbooks, as observed by an enumerator (data collector), with significant variation across contexts, ranging from under 20 percent in the Central African Republic, Madagascar, and Niger to more than 90 percent in Bangladesh, Jordan, Montenegro, and Pakistan's provinces of ICT, KP, and Punjab.
- **SCHOOL MANAGEMENT.** Effective school management plays a critical role in cultivating a productive learning environment. In fact, the GEPD data shows that school management stands out as a key differentiator of top performing schools. There are substantial differences in the presence and quality of core operational functions across the best and worst performing schools. Furthermore, principals (headteachers) often lack an accurate picture of the challenges they need to address. In all but three systems surveyed, principals overestimate the percentage of teachers who can correctly answer simple numeracy and literacy questions, with differences above 30 percentage points in ICT, KP, and Sindh in Pakistan.

Key Policies Affecting School-Level Practices

- **LEARNER PREPARATION.** The low levels of school readiness mentioned above partly reflect gaps in the design and implementation of policies that support early childhood development. Access to preprimary education, caregiver support, and proper nutrition are critical elements of early childhood development. While most systems surveyed have relevant early childhood policies in place, implementation often falls short. For example, a majority of education systems had in place policies guaranteeing preprimary education at the time of the survey. Yet secondary data sources indicate that in most systems less than 40 percent of children ages 36–59 months attend an early childhood education program.^{vi} A key challenge is ensuring that policies on paper translate into effective practices on the ground.

^{vi} Data on children ages 36–59 months who attend an early childhood education program are not available for the following countries that guarantee preprimary education: Chad, Ethiopia, Pakistan's ICT.

- **TEACHER EFFECTIVENESS.** Similarly, while policy frameworks for teacher development are **comprehensive on paper, their implementation frequently falls short.** Most systems have policies for in-service training,³ yet recent participation by teachers in such training vary widely, from four percent in Balochistan (Pakistan) to 72 percent in Peru. Moreover, research shows that even where training occurs, it tends to be brief, overly theoretical, and disconnected from classroom practice.⁴ To be effective, training must be tailored to teachers' needs, integrated with classroom practice, and reinforced through effective instructional leadership practices, including coaching, classroom observations, and provision of actionable feedback.⁵ These are areas where GEPD data show significant room for improvement. While classroom observations are relatively common, with at least 80 percent of teachers reporting having their classroom observed recently across close to half the systems surveyed (ranging from 40 percent in Sindh to 99 percent in Edo State), the provision of feedback is often limited. In only three systems (Bangladesh, Edo State, and Montenegro), at least 80 percent of teachers report receiving feedback following an observation. Similarly, across all systems except one, less than 80 percent of the teachers had lesson plans, with rates ranging from 27 percent in Balochistan to 86 percent in Jordan.
- **INPUTS AND INFRASTRUCTURE.** While policies related to basic inputs and infrastructure such as textbooks, water, and toilets exist on paper across most systems surveyed, **access to basic inputs and infrastructure is often impacted by poor implementation.** The result is that *de jure* policies do not translate into textbooks in classrooms or running water and functioning toilets in schools. Where policy frameworks are notably absent, however, is in education technology. Only four surveyed systems (Jordan, KP, Montenegro, and Rwanda) had policies governing access to education technology at the time of survey implementation. Given the increasingly important role of technology-enhanced teaching and learning (for example, through adaptive learning software or AI-assisted tutoring⁶), putting in place appropriate policy and regulatory frameworks to guide technology-aided education is an important first step.
- **SCHOOL MANAGEMENT.** Effective school management requires continuously supporting principals. Yet across surveyed systems, **the support structures to help principals develop into effective leaders are largely missing.** Access to mentoring or coaching from experienced principals remains scarce, with no education system offering such opportunities to more than half of its principals. This is a missed opportunity, as targeted training programs can improve both the performance of the principal and student learning.⁷

Politics and Bureaucratic Capacity

- **Behind every policy gap is a political and bureaucratic context that shapes whether reforms succeed or fail.** The GEPD goes beyond schools and classrooms to measure this, drawing on interviews with more than 3,500 public officials. What emerges is a story of both shared structural weaknesses and large variation across systems: high leadership turnover, weak accountability, and officials who are often disconnected from frontline realities are recurring themes, but the extent of each challenge differs across systems.
- **Education officials, who design policies, allocate resources, and manage personnel, are largely committed to their work.** Eighty-one percent report being satisfied working in government, and most want to remain in public service despite believing they could earn more in the private sector. This commitment represents an important, and often underutilized, asset for strengthening education systems.
- **This commitment rests on precarious foundations and is challenged by high leadership churn.** Across all levels, education systems surveyed have experienced frequent leadership turnover, with Peru representing an extreme case with 12 different ministers over a six-year period. Similarly, more

than half of all department heads across surveyed systems were replaced within the year leading up to the GEPD implementation. When leadership frequently changes, institutional memory erodes and long-term reforms lose momentum.

- **Political interference undermines merit-based management.** Overall, one in four public officials report that hiring and promotion decisions favor political power over merit. Around 30 percent of all public officials believe that serious misconduct—including distorting procurement processes or hiring teachers based on favoritism or reporting false information—is likely to go unpunished. These practices undermine the ability of systems to select and reward qualified personnel and to design effective, evidence-based policies.
- **Even motivated officials struggle to advance learning goals without data or incentives.** Most systems surveyed have adopted national learning goals—a sign that political commitment to better education outcomes is growing. Yet having goals on paper is not enough. About half of officials indicate that effective information management systems are absent, while fewer than half say their organizations recognize or reward efforts toward national learning objectives. Systems that cannot track whether or not learning improves have no basis to reward those responsible for driving improvements to meet learning targets.
- **Officials are often disconnected from the realities they are meant to manage.** Across systems, public officials estimate that the typical classroom has 44 students, while school observations reveal the actual figure is 24. Similarly, officials believe that teacher absenteeism stands at 11 percent, while actual measurements show rates of 22 percent. These are not minor discrepancies. When the people designing and implementing staffing policies, setting class size norms, and allocating resources have a distorted picture of what happens on the ground, the resulting policies are likely to miss the mark.

From Diagnosis to Action

The specific bottlenecks to service delivery differ across education systems, but the process for identifying and addressing them is the same: systematic diagnosis based on reliable data, evidence-based action, and ongoing monitoring. This is why it is important for education systems to invest in improving data systems that can help identify the biggest challenges to improving learning outcomes. Insights from system-level data, such as those provided by the GEPD, can be a useful starting point for helping countries diagnose bottlenecks and implement actions to address them.

The GEPD also offers a suite of global public goods. These include open-source survey instruments, a comparable cross-country database on the drivers of learning, and resources that can enable countries to implement the GEPD independently or incorporate its modules into national monitoring and evaluation systems.

Evidence from early adopters shows that GEPD data have proved valuable not only for diagnosis but also for action. The data have informed Country Partnership Frameworks, Public Expenditure Reviews, Global Partnership for Education (GPE) country compacts, country-level diagnostic reports, and project designs. Several countries are conducting second rounds of the GEPD, including Ethiopia and Peru, to track changes over time and monitor progress.

The 2025 World Bank Group Education and Skills Strategy calls for replicating and scaling what works. Doing that requires identifying both opportunities and challenges within education systems, which is exactly what the GEPD is designed to do. The key actions below offer a starting point for countries looking to move from diagnosis to action.

Table ES1: Key actions for countries implementing the GEPD

Short-term	• Secure political and technical ownership of the initiative, and designate a champion in the government. • Ensure high quality data collection (for example, rigorously train enumerators, conduct real-time data checks, and compare results against other data sources). • Build understanding of the results among key counterparts through technical workshops.
Medium-term	• Prioritize reform areas based on the largest service delivery bottlenecks identified by the data. • Institutionalize monitoring of key indicators (for example, adding relevant GEPD measures into EMIS, annual sector reviews, or district-level dashboards). • Repeat and/or adapt GEPD modules to conduct follow-up rounds to track progress.
Long-term	• Use data to incentivize performance and strengthen accountability mechanisms. • Integrate data systems (for example, linking GEPD data with other national and international surveys of service delivery and learning outcomes). • Build analytical capacity within the government to allow for sustainable data collection and analysis to inform reform priorities.

Looking Ahead: Extending The Diagnostic Agenda Beyond Primary Education

As countries seek to build the skilled workforce their economies need, understanding how secondary, Technical and Vocational Education and Training (TVET), and higher education systems develop job-relevant skills is critical. Countries already draw on a range of tools to monitor performance across these subsectors, including Education Management Information Systems (EMIS), learning assessments, teacher management systems, and skill-specific instruments, such as the Skills Toward Employment and Productivity (STEP) Skills Measurement survey and the IFC's Vitae tool.^{vii} What is currently missing, however, is an integrated diagnostic that brings together service delivery, *de jure* and *de facto* policy coherence, and political and bureaucratic dynamics for education beyond the primary level, as done by the GEPD.

For an expanded diagnostic tool, the GEPD offers a strong foundation to build on. The Survey of Public Officials, which assesses bureaucratic capacity and political context and the Policy Survey, which assesses *de jure* policies, can be adapted and expanded to cover higher levels of education, as well as private-sector engagement and financing. Countries looking for a deeper diagnostic on service delivery and *de facto* policy implementation can utilize an adapted version of the School Survey to cover higher levels of education. Experience with GEPD implementations also offers practical lessons for future diagnostics: while the GEPD has already reduced data collection costs by over 50 percent compared to earlier tools, further efficiencies could be achieved through greater use of technology and a more modular survey design.

It is critical to build on what exists, and extend the GEPD's tried and tested approach to where it is most needed. Even as the GEPD evolves to meet the demands of the skills-for-jobs agenda, its core instruments remain essential for what is most urgent: helping education systems diagnose problems, implement solutions, and ensure that every child who enters school leaves with the skills to thrive.

^{vii} See further information on the World Bank STEP Skills Measurement Program here: <https://microdata.worldbank.org/collections/step>, and IFC's Vitae tool here: <https://www.vitaeready.org/benchmarking-data/> (Retrieved February 10, 2026).

INTRODUCTION

From Classrooms to Careers—Why Foundational Skills Matter

Fatima sits at her desk in a rural primary school in central Punjab, staring at a page of text she cannot understand. She dreams of becoming a doctor one day and working in the local health clinic in her neighborhood. Yet despite being in fourth grade, she struggles to read even simple sentences. Around her, different actors in the education system are all working on what they believe will help Fatima and children like her:

- Her teacher, Mr. Usman, received a new reading curriculum last month from the Ministry of Education. With 65 students in his classroom and minimal training in teaching phonics, he is racing to cover the required material before year-end examinations.
- The district education officer recently completed a school inspection, primarily checking attendance records and school infrastructure. She noted the broken water pump but had no mechanism to record or address concerns about the lack of reading proficiency among Grade 4 students.
- At the regional education office, officials are implementing a new tablet-based program they believe will transform learning, though teachers have received no training on integrating technology into instruction. In any case, electricity shortages are common in this part of the country, raising questions about the feasibility of the program.
- Meanwhile, an international NGO has launched a teacher training initiative focused on phonics instruction, but it operates independently from the government's professional development system.
- Back in the capital, policy makers debate a new education reform focusing on expanding access to secondary education, even though learning outcomes at the primary level remain alarmingly low.

In this fragmented landscape, Fatima's chance of acquiring basic literacy and numeracy skills, let alone acquiring the specialized skills needed to become a doctor one day, remains a distant dream. Each stakeholder is working on their assigned piece of the puzzle, but no one sees the complete picture of the education system and understands how its components interact. As a result, this fragmented approach means that despite the best intentions of all actors, the education system fails to deliver its most fundamental promise: ensuring that all children learn.

Fatima's story is one of millions playing out daily across the globe. The learning crisis it represents is not just about individual students, teachers, schools, or even specific interventions. It reflects the complex interplay between politics, policies, and practices that create the conditions in which learning either flourishes or falters.

Countries have achieved remarkable success in expanding access to schooling. While Fatima and millions like her may struggle to learn in classrooms, the simple fact that they are able to attend school at all represents a significant global achievement. Since the 1970s, primary school enrollment has surged across the developing world, with remarkable progress in Sub-Saharan Africa and South Asia. The progress in girls' access to schooling has been most striking, resulting in women having greater access to schooling today than 50 years ago in every country in the world.⁸

The economic benefits of improved access to schooling are clear, with an estimated 10 percent average wage gain from an additional year of schooling.⁹ Schooling also has important social benefits, such as reduced child mortality and delayed childbirth.¹⁰ Despite these achievements, more efforts are needed to continue to expand access, as an estimated 250 million children remain out of school.¹¹ Exclusions persist for the most marginalized segments of the population, particularly those affected by poverty, disability, and conflict.

Unfortunately, progress in expanding schooling has not translated into equivalent learning gains. Despite spending years in classrooms, millions of students fail to master even basic literacy and numeracy skills. Across 48 low- and middle-income countries, most students in the first three instructional years are not acquiring basic literacy subskills, such as the ability to decode printed words fluently or to master the names and sounds associated with letters.¹² The latest available simulated estimates suggest that Learning Poverty in low- and middle-income countries has surged to an estimated 70 percent. These statistics highlight the main challenge facing global education today: having succeeded in getting children into schools, how can we ensure that time spent in school results in learning gains?

The learning crisis affects not only foundational skills: it undermines the core infrastructure that drives skills development for more and better jobs. Without foundational skills, millions of young people enter the workforce fundamentally unprepared for the demands of the labor market. Employers worldwide report growing skills gaps, with an estimated 63 percent of employers citing skills shortages as a major barrier to business transformation and growth.¹³ One estimate suggests that as many as 85 million jobs may go unfilled by 2030 for lack of candidates with the right skills.¹⁴ At the same time, technology is reshaping labor markets and the skills landscape at a rapid pace, with projections indicating that up to 60 percent of workers will require retraining by 2030.¹⁵ Without foundational skills, individuals are less able to adapt to the changing skills needs in a rapidly evolving labor market.

For individuals, the consequences are stark: those without foundational skills earn significantly less throughout their lives, ultimately entrenching intergenerational poverty and undermining economic growth. A one standard deviation increase in literacy skills is associated with a 15 percent increase in earnings globally.¹⁶ Other estimates suggest earning gains of 20 percent from a one-standard-deviation increase in numeracy skills.¹⁷ Recent estimates from Indonesia suggest an 11 percent increase in adult earnings associated with a one-standard-deviation increase in foundational literacy and numeracy skills.¹⁸ Meanwhile, the intergenerational consequences of failing to acquire foundational skills are severe, as lower education attainment perpetuates cycles of poverty and limits social mobility. The economic impacts of weak foundational skills extend far beyond individual paychecks and spill over through entire economies by reducing productivity, constraining innovation, and limiting growth.

Behind the learning crisis lies a disconnect between inputs and outcomes. More resources do not automatically produce better learning. In fact, education systems are shaped by interacting networks of relationships, incentives, and constraints at multiple levels. What teachers do in classrooms is shaped by the quality of their training, the expectations set by the curriculum, the support they receive from school leaders, and the accountability structures established through policies and regulations. School principals (headteachers) balance competing demands while navigating bureaucratic requirements and community

expectations. District officials implement policies designed at higher levels, often with limited resources and capacity. Political economy factors, from teacher union dynamics to electoral incentives, influence which reforms are prioritized and how effectively they are implemented. Even well-designed interventions can fail when they don't account for the broader system context.

Tackling the learning crisis requires understanding the education system. The World Development Report (WDR) 2018 argued that the learning crisis has multiple causes: poor service delivery in schools and communities, policies that are not aligned with learning for all, and unhealthy politics and low bureaucratic capacity. At the school level, struggling education systems typically lack one or more of four key ingredients: prepared learners who are ready to learn, quality teaching that engages students so that they acquire knowledge and skills, learning-focused inputs such as appropriate textbooks, and finally, effective management that brings all these elements together. But the problems are not just at the school level. These deficiencies are typically symptoms of deeper problems, such as misaligned policies and administrative systems that lack the incentives and capacity to implement reforms effectively.

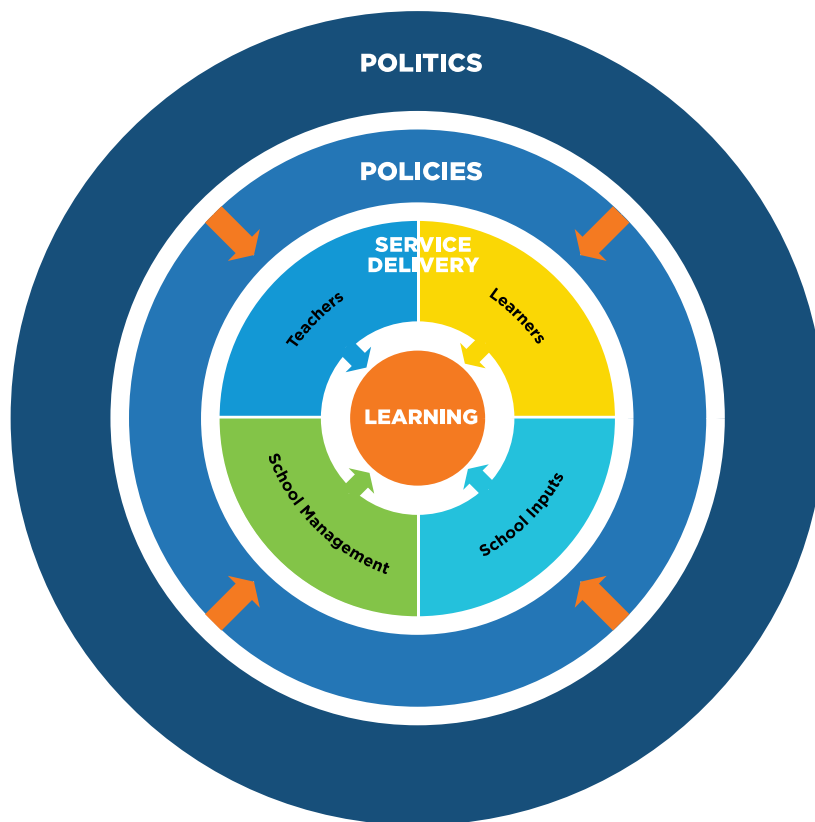
To improve learning for all children, countries need to know where they stand across the critical dimensions of service delivery, policies, and politics. This requires system-level data that provides actionable, regular, and reliable data on the key drivers of learning. Yet far too many countries lack such data. This data gap is not trivial. Without systematic information on what is happening in classrooms, whether policies are being implemented as intended, and whether the bureaucratic machinery is functioning effectively, governments are left designing reforms based on incomplete or outdated information. Many countries have data on enrollment and, increasingly, on learning outcomes. Far fewer have data on the drivers of those outcomes: whether teachers have the content knowledge to teach, whether textbooks reach classrooms, whether school leaders provide instructional support, or whether public officials have the tools and incentives to do their jobs effectively. The GEPD was designed to address this problem by providing actionable data on the underlying drivers of low learning.



The Global Education Policy Dashboard: A Tool to Measure Drivers of Learning



Figure 1.1: The GEPD framework



The Global Education Policy Dashboard provides countries with a clear picture of how their education systems are performing by measuring critical dimensions of education service delivery, policies, and politics. By tracking indicators that research has shown to be strongly associated with learning outcomes, the GEPD highlights where systems are falling short in delivering quality education, identifies gaps between current practice and what evidence suggests is most effective for promoting learning, and helps governments set priorities and track progress as they work to close those gaps.

The structure of the GEPD builds on the WDR 2018 conceptual framework (Figure 1.1). By mapping the entire education system, the GEPD offers a strong foundation for diagnosing challenges and opportunities in primary education. The tool is centered on the goal of improving foundational learning outcomes. Surrounding this central goal are four main school-level drivers of learning: prepared learners, quality teaching, adequate inputs and infrastructure, and capable school management. Struggling systems lack one or more of these critical school-level ingredients.¹⁹ The circle surrounding these proxies for the *de jure* and *de facto* policies that affect each of these school-level indicators, and the final circle captures the political context and bureaucratic capacity of the system.

The GEPD collects new data in each education system using three instruments: a School Survey, a Policy Survey, and a Survey of Public Officials (see Box 1.1). Data collection involves school visits, classroom observations, legislative reviews, teacher and student assessments, as well as interviews with teachers, principals, and public officials. In addition, the GEPD also draws on existing data sources to complement the new data collected.

Box 1.1: GEPD Survey Instruments

The School Survey^{viii} collects data primarily on school-level practices (the quality of service delivery in schools), but also on select *de facto* policy indicators. It is carried out in a nationally representative sample of schools and takes approximately four hours to complete per school. It includes eight core modules: Teacher Roster, School Information, Principal Questionnaire, Teacher Questionnaire, Teacher Assessment, Grade 1 Direct Assessment, Grade 4 Classroom Observation, and Grade 4 Student Assessment.

The Policy Survey^{ix} collects information about *de jure* education policies. This survey is completed by key informants in each country to identify key elements of policies related to teachers, school management, inputs and infrastructure, and learners. It generally takes several days to gather and analyze the relevant information.

The Survey of Public Officials^x collects information on technical and leadership skills, work environment, stakeholder engagement, impartial decision-making, and attitudes and behaviors at central, regional, and district levels. The aim is to assess bureaucratic capacity as well as political factors affecting education outcomes. The survey generally takes 30–45 minutes per public official.

Roughly 73 percent of the variation in Grade 4 student achievement is explained by drivers of learning captured through the GEPD.

The GEPD's comprehensive yet streamlined design keeps costs relatively low (less than half compared to earlier instruments) and allows for rapid data collection, while explaining a significant share of what drives learning. Roughly 73 percent of the variation in Grade 4 student achievement is explained by drivers of learning across multiple layers of the education system (school-level practice indicators, *de facto* policy indicators, and politics indicators), as shown in Figure 1.2. This finding provides empirical support for the GEPD's conceptual framework and confirms that the selected metrics capture meaningful aspects of education systems.

^{viii}The School Survey consists of streamlined versions of existing instruments used by the World Bank and other partners, including Service Delivery Indicators (SDI) Surveys on teachers and inputs/infrastructure; TEACH Primary on pedagogical practices; the Global Early Child Development Database (GECDD); Measuring Early Learning Quality and Outcomes (MELQO); Anchor Items for Measurement of Early Childhood Development (AIM-ECD) on school readiness of young children; and the Development World Management Survey (D-WMS) on management quality; along with new questions to fill gaps in these existing instruments.

^{ix}The Policy Survey draws on the SABER approach to policy data collection that the World Bank has used in the past.

^xThis Public Officials Survey is a streamlined and education-focused version of civil-servant surveys conducted in several countries by the Bureaucracy Lab, a joint initiative of the Governance Global Practice and the Development Impact Department of the World Bank.

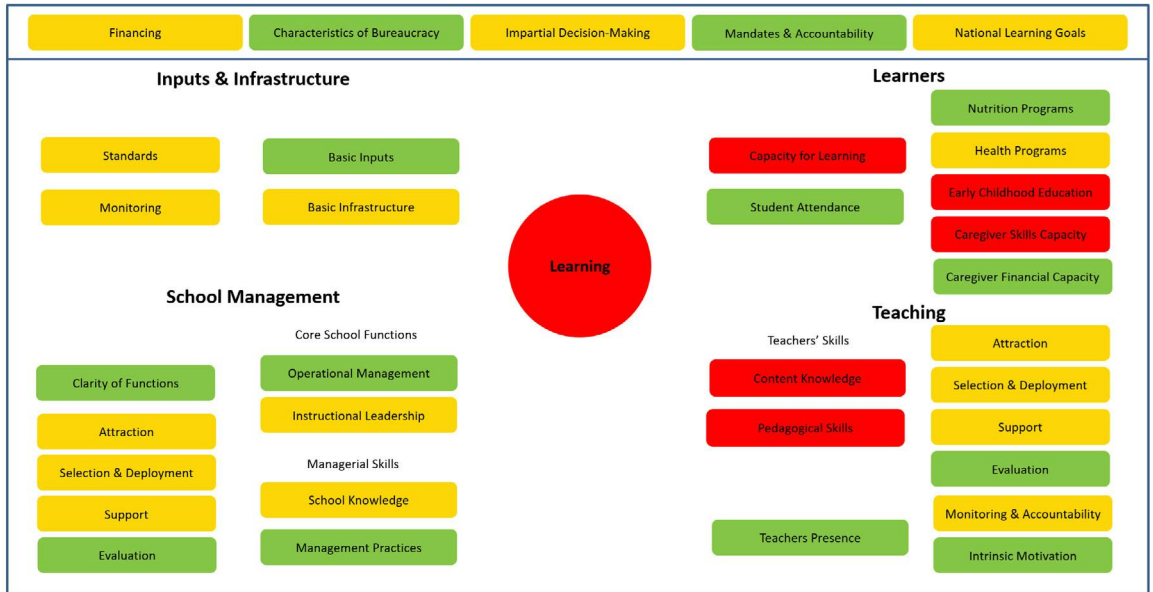
Figure 1.2: GEPD's practice, de facto policy, and politics variables explain a substantial proportion of the variation in Grade 4 student learning outcomes



Notes: Fitted values are produced using a pooled OLS regression across countries by regressing school-level Grade 4 student learning scores on GEPD school practice, de facto policy, and politics indicators. Interaction terms are included so that coefficients are allowed to vary by country. No other controls are included in the regressions. Madagascar is omitted for lack of data on student absence and TEACH Primary (a classroom observation tool). In Montenegro, politics variables are constructed as the average of responses from Ministry of Education officials. In all other systems, they are constructed as the average of responses from district-level officials corresponding to each district in the school dataset.

A core design principle underpinning the GEPD is that data should be immediately actionable. Each participating country receives its results in the form of an integrated dashboard that displays the core indicators using traffic-light color coding to signal areas of strength (green), caution (yellow), and improvement (red). Figure 1.3 illustrates this using Peru's first round GEPD results. The snapshot makes it clear that the most pressing gaps at the school level were in teacher skills, learner readiness, and principals' knowledge about their own school. Tracing these upward to the policy layer, the dashboard reveals that support for teachers and principals, along with the quality of early childhood education, were among the weakest policy areas, signaling priority areas for action. At the same time, some aspects of the bureaucratic environment, such as mandates and accountability, appeared relatively strong, suggesting levers that could help close critical policy gaps. Peru is one of several countries now conducting a second round of the GEPD to enable comparisons over time and understand whether reforms are translating into measurable progress.

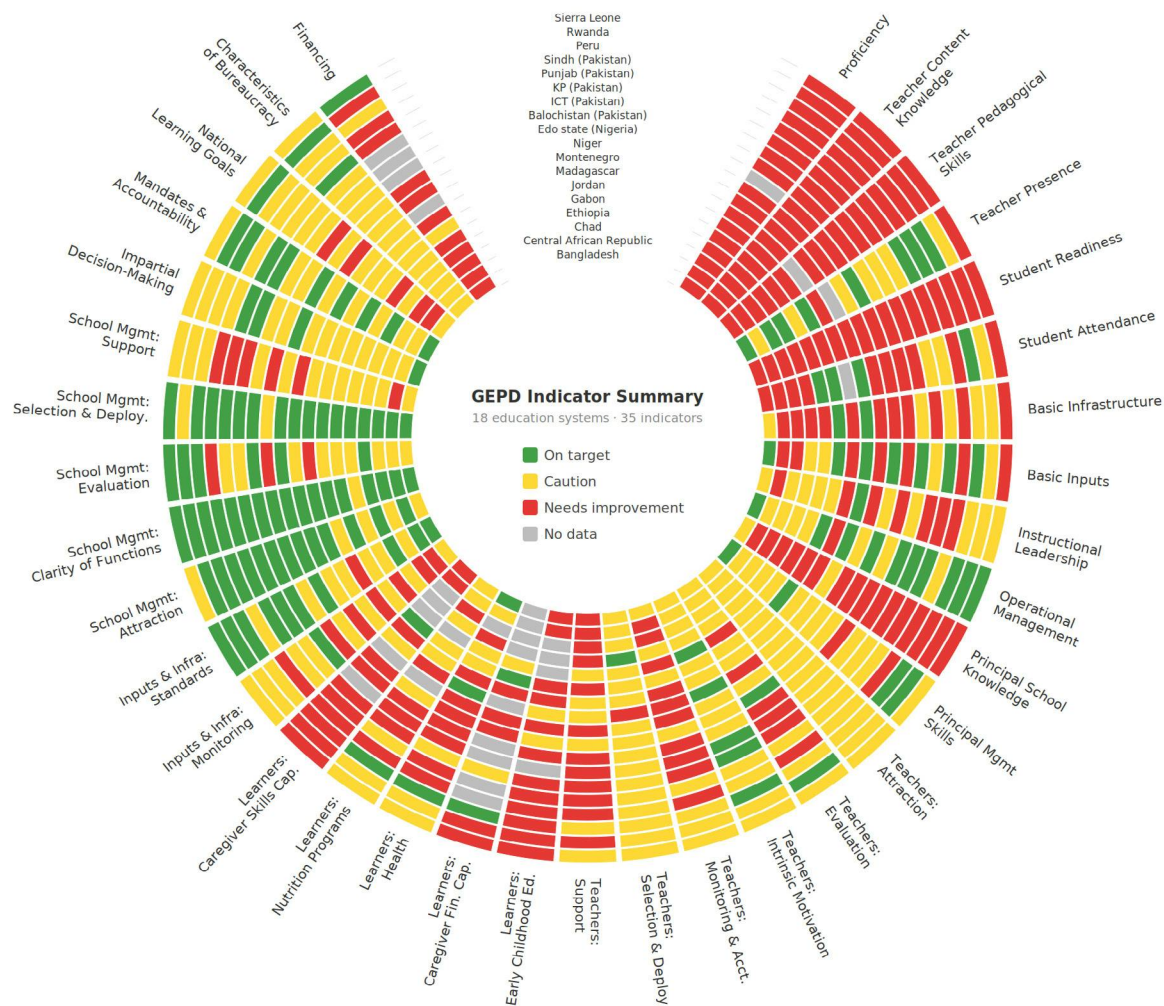
Figure 1.3: GEPD education system snapshot for Peru



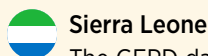
The real power of the GEPD’s framework is shown by what becomes visible when the same indicators are measured across many systems at once.

The real power of the GEPD’s framework is shown by what becomes visible when the same indicators are measured across many systems at once. Figure 1.4 brings together GEPD results for 18 education systems in a single view, with each ring representing one system and each segment representing one of the core indicators. Patterns that would be invisible in any single-country dashboard emerge clearly: some indicators show persistent red across nearly all systems (for example, teacher content knowledge, pedagogical skills, and student readiness), pointing to systemic weaknesses that transcend context. Others reveal sharp variation, identifying areas where some systems have found solutions that others have not yet adopted (for example, basic inputs and infrastructure). This cross-country lens helps demonstrate the GEPD’s value not just as a national diagnostic, but also as a global learning tool.

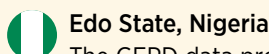
Figure 1.4: GEPD results across education systems



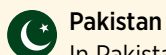
The integrated nature of the GEPD data has enabled countries to use them in diverse ways, from informing national strategies and program design to strengthening policy dialogue and identifying reform priorities (Box 1.2). For example, in Sierra Leone, data on teacher pedagogical practices informed the design of continuous professional development interventions. In Edo State, Nigeria, the data validated strengths of the EdoBEST reform while drawing attention to previously overlooked gaps in early childhood education. In Bangladesh, GEPD findings are shaping the design of the government's Primary Education Development Program, with proposed interventions targeting school readiness and teacher support. These examples confirm the value of providing policy makers with a comprehensive view of the entire education system, as the GEPD does, and better positioning them to act on the evidence.

Box 1.2: How has the GEPD data been used to date?

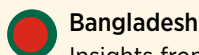
The GEPD data informed multiple high-value products, including Sierra Leone's Zero Learning Poverty Plan and National Digital Learning Strategy (via the Education Technology Readiness Index, ETRI). The data also informed interventions to improve teaching practice. For example, data on teacher pedagogical practices were used to inform design and implementation of continuous professional development interventions. Classroom videos from TEACH observation tool^{xi} have been used to facilitate “teacher circles” and discussions aimed at instructional improvement.



The GEPD data provided evidence highlighting the strengths of the government's flagship EdoBEST project, including high teacher presence and effective school management procedures. The data also helped identify key gaps within the EdoBEST reform agenda, prompting a stronger focus on early childhood education, which had not previously been a central reform area. GEPD data were used to cross-check other data sources, such as the then-newly developed Basic Education Management Information System (BEMIS), and to inform various strategic documents, including the state's plan to accelerate foundational learning.



In Pakistan, the findings sparked discussions on priority reforms and helped draw attention to persistently poor foundational learning outcomes. The School Education Department (SED) of the Government of Balochistan has leveraged GEPD insights in dialogue with key stakeholders. Based on the GEPD results, the provincial government has shown considerable interest in strengthening early childhood development and continuous teacher professional development initiatives.



Insights from the GEPD surveys have been used to inform the design of the government's Primary Education Development Program. Building on recommendations from the GEPD analysis, proposed interventions emphasize expanding school readiness programs through a phased rollout of a preprimary education initiative. Beyond early childhood education, the program also emphasizes improving teacher support through improved preservice training and continuous progressional development, with a focus on content mastery and pedagogical practices, which were areas highlighted by the GEPD data as requiring support.



The second round of data collected in Ethiopia illustrated important signs of progress since 2020 (for example, improvements in student and teacher attendance, and school readiness), but also persistent challenges (for example, low student proficiency in Grade 4). The results were used to spark a discussion among representatives from the Ministry, development partners, private sector, and philanthropic donors on how reforms to support teachers (for example, expanded professional development and accountability/performance systems) can strengthen teaching and learning. Collecting multiple rounds of comparable data, as done in Ethiopia, offer opportunities for decision-makers to analyze improvements and key bottlenecks. The demand for a second round of data collection highlights the value stakeholders get from the GEPD in generating important insights about their education systems.

^{xi} Consent for classroom video recording must be obtained in accordance with country-specific requirements.

This report draws on data from education systems^{xiii} across diverse contexts to present a comprehensive view of key bottlenecks and opportunities for progress (Table 1). Across all education systems, more than 4,000 schools were visited, more than 17,000 teachers and 3,500 public officials were interviewed, and more than 76,000 Grade 4 students and 12,000 Grade 1 students were assessed. Spanning Latin America, Sub-Saharan Africa, South Asia, the Middle East, and Europe, the countries surveyed reflect a range of income levels and governance structures. The sections that follow unpack the rich data collected by the GEPD, examining what is happening in schools and classrooms, the policies that shape these practices, and the political and bureaucratic environment in which they operate. Together, the evidence offers not just a snapshot of where education systems stand, but a tactical roadmap for prioritizing reforms to enhance learning.

Table 1: Number of schools, principals, teachers, students, and public officials surveyed

	Year	Schools	Principals Interviewed	Teachers Interviewed	Teachers Assessed	Teachers Observed	Grade 1 Students	Grade 4 Students	Public Officials
Bangladesh	2024	640	640	2,921	2,890	640	1,914	10,850	433
Chad	2023	262	262	1,101	1,086	440	782	5,426	207
Central African Republic	2023	200	200	647	804	168	604	4,562	213
Edo State (Nigeria)	2023	200	200	901	901	196	589	3,942	152
Ethiopia	2025	300	300	1,392	1,393	299	899	6,571	202
Gabon	2023	196	189	615	634	155	501	3,546	201
Jordan	2019	250	248	1,091	1,063	246	735	4,963	203
Madagascar	2021	200	197	1,234	1,212	.	596	1,777	202
Montenegro	2025	143	99	603	536	82	427	1,990	52
Mozambique	.	.	.	.	.	.	.	.	208
Niger	2022	270	270	1,058	1,106	262	691	5,220	213
Balochistan (Pakistan)	2023	200	198	638	632	190	577	2,497	148
ICT (Pakistan)	2023	100	100	432	431	100	300	2,420	47
KP (Pakistan)	2022	200	200	628	629	197	593	3,797	139
Punjab (Pakistan)	2023	200	200	860	860	200	597	3,972	177
Sindh (Pakistan)	2023	200	199	658	592	185	552	2,400	152
Peru	2019	206	204	760	760	189	605	2,666	203
Rwanda	2020	200	199	951	945	194	600	5,259	207
Sierra Leone	2022	302	301	1,114	1,115	518	917	5,124	207
Total		4,269	4,206	17,604	17,589	4,261	12,479	76,982	3,566

^{xiii} The 19 education systems used for the analysis in this report include (in chronological order of implementations): Peru (1st round), Rwanda, Jordan (1st round), Madagascar, Mozambique (partial), Edo State (Nigeria), Sindh (Pakistan), Punjab (Pakistan), KP (Pakistan), Islamabad CT (Pakistan), Balochistan (Pakistan), Niger, Gabon, Chad, Sierra Leone, Bangladesh, Central African Republic, Ethiopia (2nd round), and Montenegro. For Madagascar, no TEACH Primary data is available. Madagascar's results are not directly comparable due to differences in the implementation approach. For Mozambique, the only available data are from the public officials survey.

2

Unpacking Learning and its Drivers— A Peek Inside 4,000 Classrooms

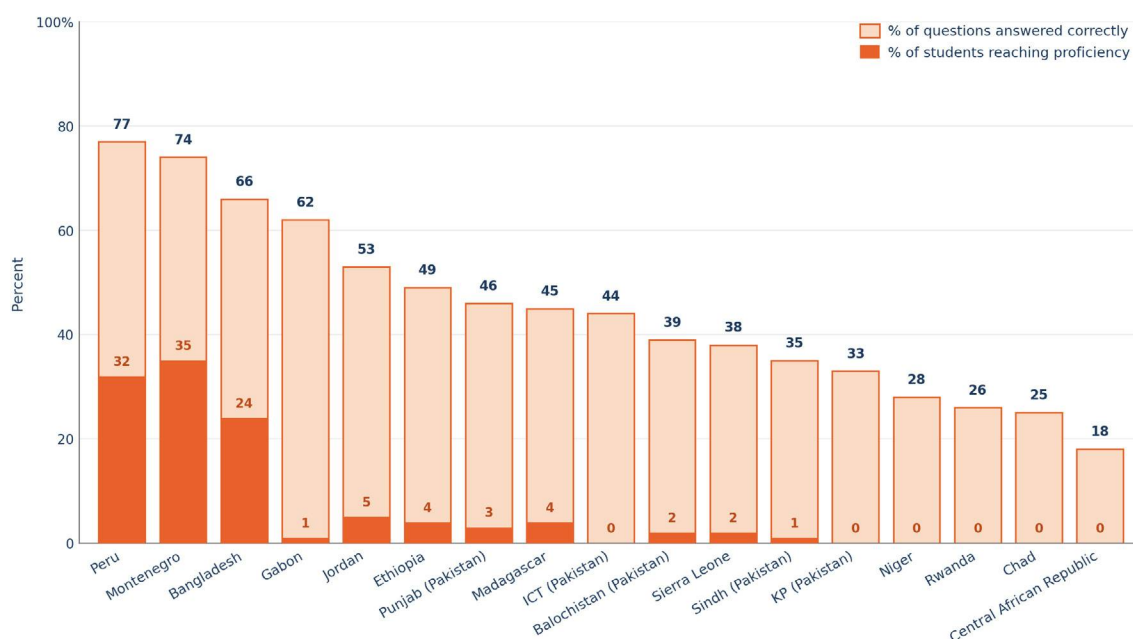


For education to lead to economic opportunity, time spent in school must translate to learning gains.

Getting children into school was the defining achievement of the past half century. Enrollment has expanded dramatically and gender gaps in access have narrowed substantially.²⁰ But access alone does not guarantee learning. Research shows that cognitive skills, rather than years of schooling alone, have powerful economic effects.²¹

Far too many students lack foundational skills in literacy and numeracy despite having spent years in primary school. The GEPD data show that in no system surveyed do children assessed through basic literacy and numeracy assessments manage to answer more than 80 percent of the questions correctly.^{xiii} In fact, across the majority of systems (14 out of 17), less than 10 percent of Grade 4 students have mastery over the literacy and numeracy skills expected at this level.^{xiv} Even in relatively better performing systems like Bangladesh, Peru, and Montenegro, roughly two-thirds of students struggle to master basic literacy and numeracy tasks (Figure 2.1a). These results are consistent with the high rates of Learning Poverty documented across many of the surveyed systems. In Chad, Ethiopia, Madagascar, and Niger, more than 90 percent of children are unable to read with comprehension by the end of primary school.

Figure 2.1a: Percentage of questions answered correctly and share of students proficient in the Grade 4 assessment

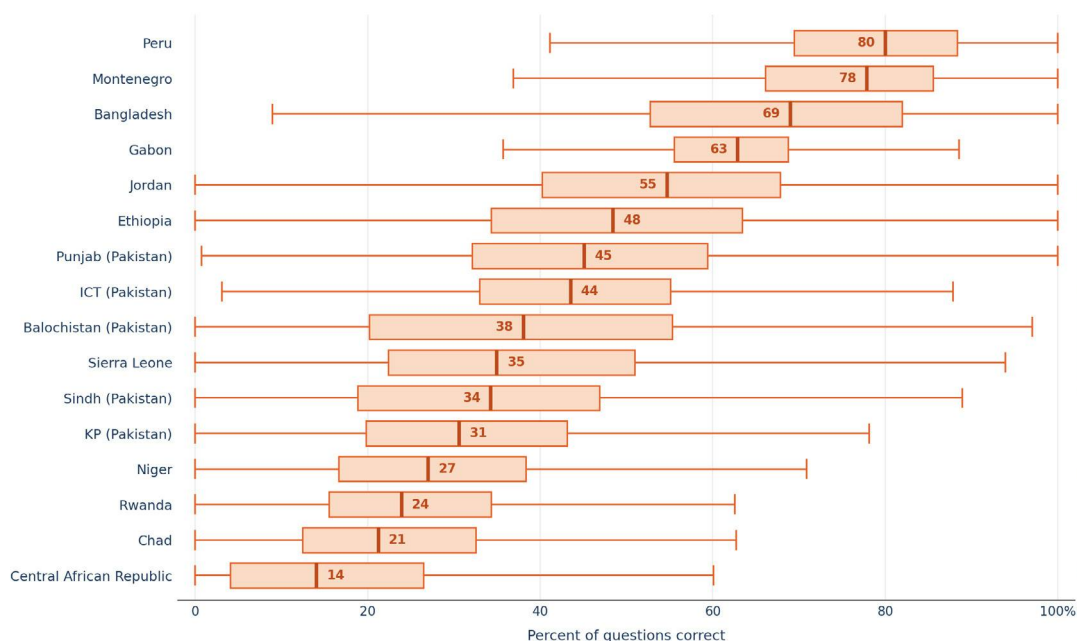


Notes: “% of questions answered correctly” represents the mean percentage of correct responses. “% of students reaching proficiency” corresponds to the percentage of children who correctly answer at least 34 out of 41 questions for the overall assessment, 14 out of 17 questions for the math assessment, and 20 out of 24 questions for the literacy assessment. The results for Edo State (Nigeria) are excluded from the graph and corresponding statistics. In Peru, the literacy assessment comprised a limited number of questions compared to the assessment used in other systems.

^{xiii} The literacy assessment covers questions related to letter identification, word recognition, and reading comprehension. The numeracy assessment covers questions designed to assess number sense, arithmetic, word problems, and sequences.

^{xiv} The threshold is set at 34 correct answers (out of 41 questions). Further details about the thresholds used in the GEPD assessments can be accessed here: <https://www.educationpolicydashboard.org/sites/default/files/resources-documents/GEPD%20Technical%20Note.pdf>.

Figure 2.1b: Distribution of percentage of questions answered correctly in the Grade 4 assessment

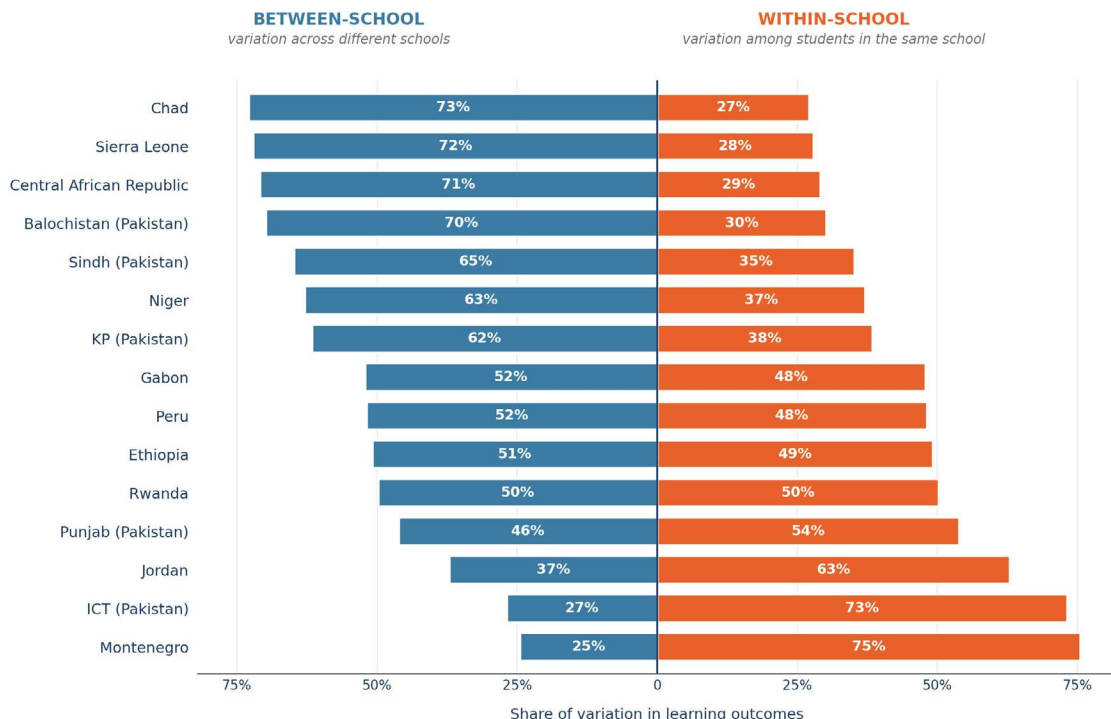


Notes: The results for Edo State (Nigeria) and Madagascar are excluded from the graph and corresponding statistics. The left edge of each box represents the 25th percentile of percent of correct answers, and the right edge represents the 75th percentile. The interquartile range is the difference between the 75th and 25th quartiles. The left whisker represents the lower adjacent value, and it is equal to the 25th percentile minus 1.5 times the interquartile range. The right whisker represents the upper adjacent value, and it is equal to the 75th percentile plus 1.5 times the interquartile range. The values below the lower adjacent value or above the upper adjacent value are excluded.

Children across most systems acquire relatively simpler literacy and numeracy subskills but struggle with more advanced foundational ones. In around half of the systems (eight out of 17), more than 80 percent of students correctly solve single-digit addition problems such as $8+7$, but only 36 percent of the students in those eight systems can correctly solve one-digit division problems such as $75/5$. Similarly, in more than half the systems (10 out of 17), more than 80 percent of Grade 4 students are able to correctly match all basic words with their corresponding images, while 72 percent of students in those ten systems can answer a basic question about a simple story. These estimates indicate that children are acquiring some basic building blocks, but failing to acquire the sequence of subskills need to read with understanding and do basic math. A sequenced acquisition of basic literacy²² and numeracy subskills is necessary to acquire more advanced foundational skills.

National averages obscure the wide variation in student performance seen within many systems. The distribution of Grade 4 scores reveals substantial differences among typical students within the same country, in both relatively high-performing systems such as Bangladesh and lower-performing systems like Balochistan (Pakistan) (Figure 2.1b). There is also considerable variation across regions within the same country, and Pakistan illustrates this phenomenon. Pakistan's ICT exhibits relatively tight clustering of typical Grade 4 scores, while Balochistan and Sindh show much broader spreads, indicating far greater heterogeneity among students in schools under the same national education framework and subnational management. Averages can also mislead in a different way. Bangladesh and Gabon have similar average scores, but different proficiency rates, because the underlying distributions of student performance differ.

Figure 2.1c: Within- and between-school variation in Grade 4 learning outcomes



Notes: For each country, a mixed-effects model with a random intercept at the school level is estimated. Students are clustered within schools, and no additional covariates are included in the model. Sampling weights are applied at both the student and school levels to account for the sampling design. The results for Edo State (Nigeria) and Madagascar are excluded from the graph and corresponding statistics.

In some systems, school quality varies widely, while in others, most of the differences in learning outcomes are inside the same classroom. Decomposing the variation in Grade 4 learning outcomes into key components (differences between schools and differences within schools) reveals stark variation across systems (Figure 2.1c). For example, in Chad and Sierra Leone, most of the variation in learning outcomes is between schools. The school a child attends is a major determinant of what that child learns because school quality varies widely. On the other hand, in Montenegro and Pakistan's ICT, the pattern reverses. Schools are more similar to each other on average, but classrooms contain students at very different learning levels, with most variation seen within schools. Pakistan's provinces again illustrate this contrast. Balochistan and Sindh show large between-school gaps. Punjab and especially ICT show larger within-school differences. These patterns have policy implications. Systems with high between-school variation need to prioritize raising performance in the lowest-performing schools. Whereas systems with marked within-school variation can benefit from prioritizing targeted instruction and differentiated support to meet students' individual needs.

Student learning outcomes are shaped by the practices, policies, and politics underpinning the education system. At the school level, several critical ingredients influence whether children learn. These include whether students arrive prepared for primary school, whether teachers have the content knowledge and pedagogical skills to teach effectively, whether classrooms and schools have the inputs and infrastructure needed for effective instruction, and whether school leaders have the knowledge and capacity to effectively manage their schools, and bring together all the elements described earlier. But we know that these school-level factors do not operate in a vacuum. They are shaped by the broader policy environment

governing aspects such as early childhood education, preservice and in-service teacher training requirements, textbook standards, and training of principals (headteachers), among others. Finally, the political context and bureaucratic capacity of the system affect how policies are formulated and translate into impact on the ground. The following sections unpack each of these layers, starting where learning begins: with the students and teachers who walk through the school door.

Lost Time: Student and Teacher Absenteeism Reduce Instructional Time

It is not possible to teach an empty seat or learn from an absent teacher. Yet in many places both are the norm. The time that teachers and students spend engaged in learning tasks is one of the most consistent predictors of learning outcomes.²³ Yet in 10 out of 17 systems, at least 15 percent of students are absent during an unannounced visit, with student absenteeism exceeding 30 percent in Pakistan's Sindh province and Sierra Leone.

Teachers are often absent from class during scheduled teaching time. In 12 out of 17 systems, more than 15 percent of teachers are absent from the classroom during scheduled teaching time. However, across

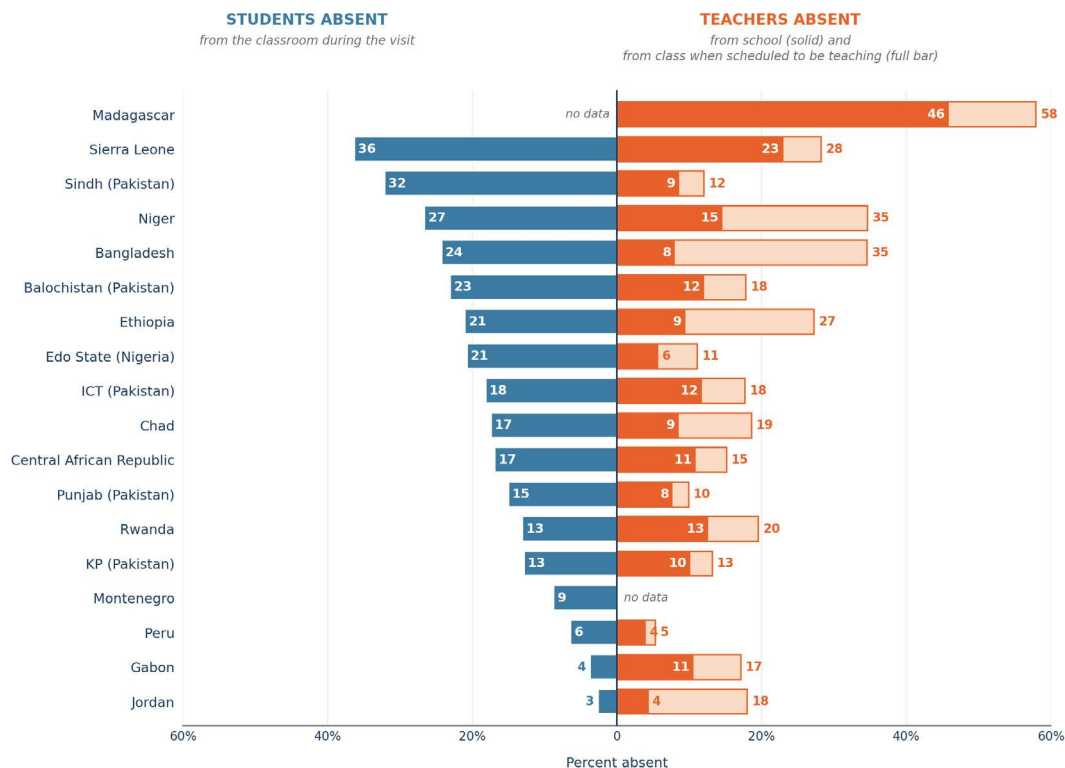
most systems, a significant proportion of teachers absent from classrooms during scheduled teaching time can be found on the school premises (Figure 2.2). This suggests that noninstructional activities, particularly administrative tasks, are pulling teachers away from scheduled teaching time. In fact, more than one in three teachers (35 percent) across surveyed systems report missing school days because of administrative processes.

Progress is possible with the right investments: In Ethiopia, school readiness improved from 33 percent to 45 percent between 2020 and 2025.

Data show that progress is possible, especially for teacher presence on school premises. For example, in Ethiopia, the proportion of teachers absent from the school premises halved between 2020 and 2025, falling from 17 percent to 9 percent. However, reducing teachers' classroom absence proved harder. The share of teachers absent from class during scheduled teaching time fell by a smaller margin, from 31 to 27 percent, underscoring that getting teachers to school is a necessary but not sufficient step toward getting them in front of students and optimizing instructional time.

The GEPD data suggest concrete entry points for reducing student and teacher absenteeism. The drivers of student absenteeism vary by context and may include the opportunity cost of attending school (including paid work, household chores, or caregiving responsibilities), while in others, distance and safety concerns may play a larger role. Evidence from programs in countries such as Mozambique suggests that providing parents with information about their children's school attendance can be effective at reducing student absenteeism.²⁴ On the other hand, frequent school monitoring and offering financial incentives based on attendance (not just increasing salaries alone) have shown to be effective for reducing teacher absenteeism.²⁵ In systems where more than one-quarter of the teachers are absent from class during scheduled teaching time, such as Niger and Bangladesh, strengthening monitoring systems, linking incentives to attendance, and reducing the administrative burden that pulls teachers out of classrooms during scheduled teaching time could make a measurable difference.²⁶

Figure 2.2: Share of students and teachers absent during an unannounced visit



Notes: Student absence and teacher absence rates are not reported for Madagascar and Montenegro respectively.

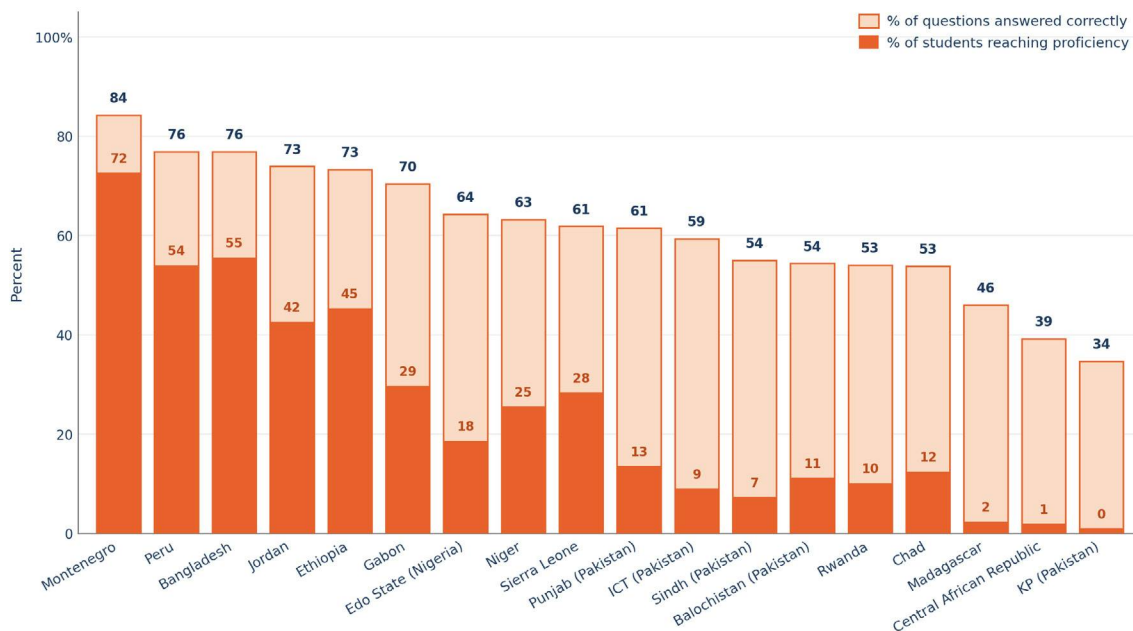
Before Day One: Many Learners Arrive at Primary School Without Basic Readiness Skills

Student readiness at the start of primary school is one of the strongest predictors of later learning outcomes. Children who enter Grade 1 with early numeracy, literacy, executive functioning, and socio-emotional skills are better positioned to keep learning in higher grades.²⁷ Socio-emotional skills acquired in early childhood, such as perseverance, motivation, and self-control, matter not just for school performance but for success in the workplace decades later.²⁸ The returns to investing in early childhood development are among the highest in education, both in terms of learning gains and long-term economic outcomes.²⁹

Students across most systems do not arrive at school with the skills needed to thrive in primary grades. The GEPD measures school readiness through a Grade 1 assessment covering early numeracy and literacy, executive functioning, and socio-emotional skills (Figure 2.3). The assessment covers skills such as recognizing single-digit numbers and simple words, and demonstrating empathy and following instructions. Only in three out of 18 systems (Bangladesh, Montenegro, and Peru) do more than 50 percent of the students demonstrate proficiency in the Grade 1 assessment, defined as answering around 80 percent of the questions correctly. Montenegro stands out as the only system where more than seven in 10 students demonstrate skills needed to thrive in primary school.^{xv} In five systems, readiness rates are in single digits.

^{xv} Students have to answer at least 80 percent of the questions on the Grade 1 assessment correctly to be considered proficient.

Figure 2.3: Percentage of questions answered correctly and students proficient in the Grade 1 assessment



Notes: “% of questions answered correctly” represents the mean percentage of correct responses. “% of students reaching proficiency” corresponds to the percentage of children answering at least 80 percent of all questions correctly.

But the data also show that progress is possible. For example, in Ethiopia, school readiness improved by more than 10 percentage points between 2020 and 2025, demonstrating that substantial gains in percentage of students achieving proficiency are achievable within a few years with the right investments.

Across all systems surveyed, numeracy is a relatively bright spot. Students across most systems answer at least 80 percent of the numeracy questions correctly, ranging from 54 percent in Pakistan’s KP (Khyber Pakhtunkhwa) province to over 90 percent in Bangladesh, Ethiopia, Jordan, and Montenegro. In 17 out of 18 systems, more than 70 percent of students answered correctly when asked which number is greater (“three or five?”), with nearly universal understanding in Bangladesh, Ethiopia, Jordan, Montenegro, and Peru. Numerical concepts are often more embedded in daily life and play, which may aid their acquisition. Acquisition of literacy can be more challenging, as it also depends on exposure to print and home literacy environment. While the GEPD assessments are not benchmarked for direct comparisons across their domains, a similar pattern of literacy emerging as the harder skill is observed in studies of early learning in Ethiopia, Nigeria, and Pakistan’s KP province.³⁰ In education systems such as the Central African Republic, Madagascar, Pakistan’s KP province, and Rwanda, less than 20 percent of Grade 1 learners were able to read the word “dog”. Socio-emotional skills lag furthest behind. In no system surveyed do students answer more than 80 percent of questions on the socio-emotional component correctly, with results ranging from 15 percent in KP to 74 percent in Peru. Socio-emotional skills, including the ability to demonstrate empathy and resolve conflict, are critical for managing relationships, emotions, and ultimately for success in the workplace.

Access to high-quality preprimary education, caregiver support, and nutrition play a key role in early childhood development.³¹ While many policies for promoting early childhood development are in place, their effective implementation is often lacking. Fifteen out of 18 systems^{xvi} had in place policies guaranteeing preprimary education at the time of the GEPD survey. Yet in most systems with policies guaranteeing preprimary education,^{xvii} fewer than 40 percent of children ages 36–59 months attend an early childhood education program. Participation ranges from a high of 64 percent in Edo State (Nigeria) to single digits in the Central African Republic, Niger, and Pakistan’s provinces of Balochistan and KP. Similarly, across 10 out of 18 systems, governments had programs to share good parenting practices with caregivers, while 16 systems maintained at least one publicly supported channel^{xviii} to promote early childhood stimulation. Yet despite institutional efforts to support caregivers, in most systems, less than 50 percent of children aged 24 to 59 months recently engaged in early stimulation and responsive care activities with any adult in the household. This engagement is around 25 percent in the Central African Republic, Madagascar, and Sierra Leone, and reaches 90 percent in Jordan and Montenegro.^{xix} Finally, eight out of 18 systems had publicly funded school feeding policies in place. Yet among all countries with a school-feeding policy, only 19 percent of schools report having a public school-feeding program, with coverage in single digits in Edo State (Nigeria), the Central African Republic, and Chad.^{xx} The GEPD data helps each system identify where to focus efforts to boost early childhood development based on their unique context. Moreover, the data highlights critical implementation gaps that prevent well-meaning policies on preprimary, parenting, and nutrition programs from translating into improvements in early childhood outcomes.

The Teaching Gap: Teachers Often Lack the Content Knowledge and Pedagogy to Teach Effectively

Teaching quality is one of the most important determinants of student learning.³² Going from low-performing to high-performing teaching can improve student outcomes by the equivalent of multiple years of schooling.³³ This is why many of the most effective foundational learning programs in developing countries, including Ghana, India, Kenya, Pakistan, and Senegal, focus on interventions that help teachers deliver effective instruction in the classroom.³⁴ Teachers’ content knowledge of the subjects they teach and their ability to engage students with that content through effective pedagogy are two critical aspects of effective teaching. Observable characteristics like qualifications and experience show a surprisingly weak link to student learning, not because they are unimportant, but because their quality is often too low to make a difference.³⁵

Far too many teachers struggle with content they are expected to teach as part of the Grade 4 curriculum. To measure teachers’ mastery of the content they teach, the GEPD asks teachers to grade mock student tests in language or mathematics, an approach mirroring their routine activities that avoids any impression of undermining their professionalism. The results are sobering. Across most systems surveyed (12 out of 18), less than 20 percent of teachers demonstrate proficiency in the subjects they teach (Figure 2.4a), defined as answering at least 80 percent of the questions correctly. This is important because prior research shows that disparities in teacher content knowledge explain approximately a fifth of the learning gap between the lowest- and highest-performing countries.³⁶

^{xvi} Data from the 2020/2021 implementation in Ethiopia are included for this estimate and other *de jure* statistics due to unavailability of policy survey data from the 2025 implementation.

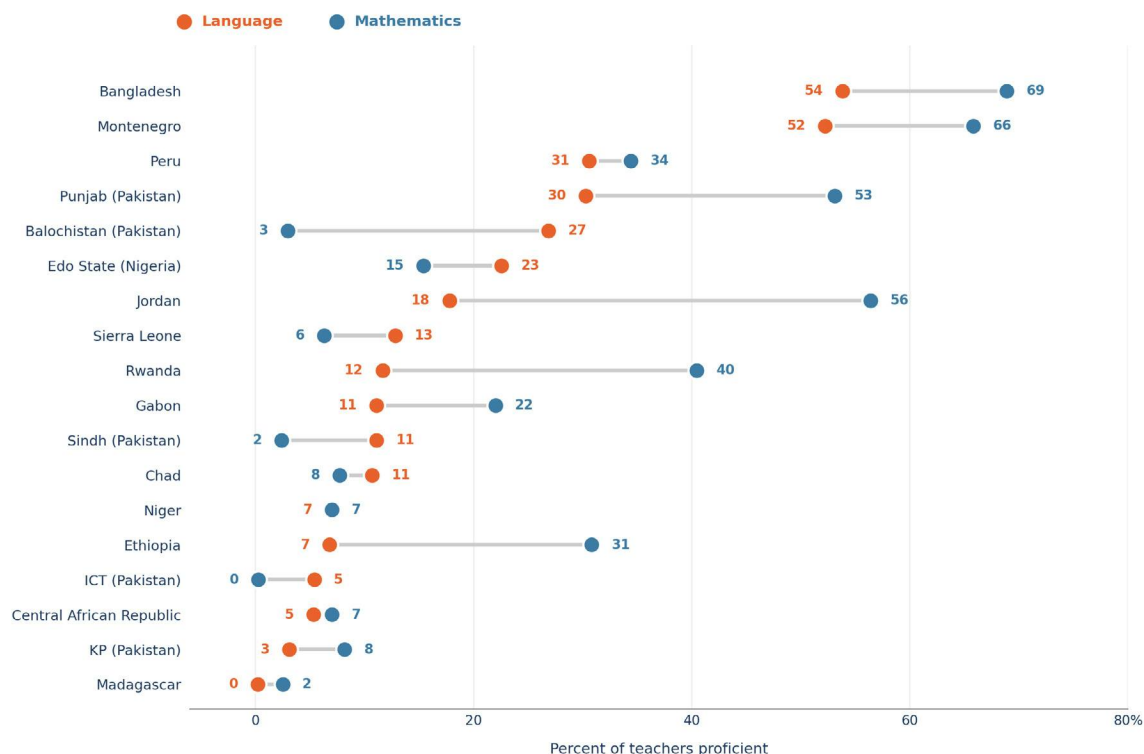
^{xvii} This refers to 11 out of 12 systems. Data on children ages 36–59 months who attend an early childhood education program are not available for the following countries that guarantee preprimary education: Chad, Ethiopia, and Pakistan’s ICT.

^{xviii} Examples of publicly supported channels include home visits, group sessions, community health programs, health center waiting rooms, school-based rooms, and mass media or information campaigns.

^{xix} Data on percentage of children ages 24–59 months who were engaged recently in activities to provide early stimulation are not available for Chad, Ethiopia, Gabon, Niger, and Pakistan’s ICT and KP.

^{xx} Madagascar and Peru are excluded from the statistics on the percentage of schools with a public school-feeding program due to data unavailability.

Figure 2.4a: Percent of teachers with content knowledge proficiency in the subject they teach



Notes: Proficiency is defined as the proportion of teachers achieving a score of at least 80 percent in the corresponding subject.

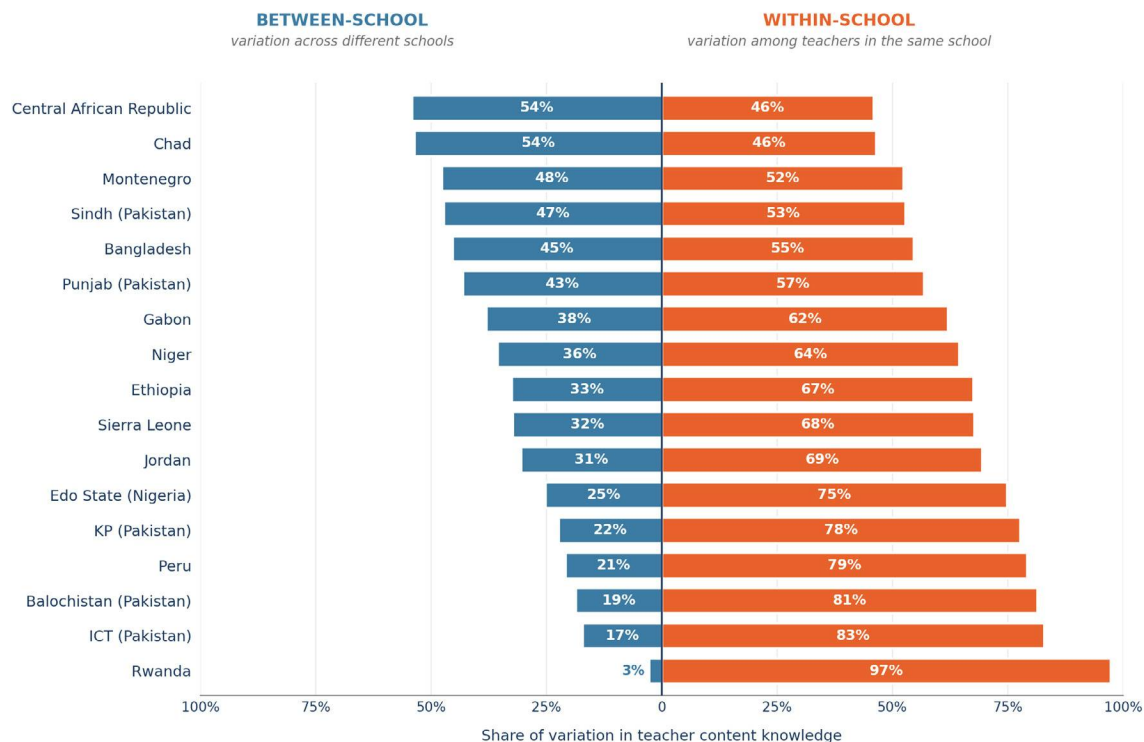
Teachers often struggle with foundational tasks, such as simple division and reading comprehension. In only five out of 18 systems, can more than 80 percent of math teachers correctly answer a simple division question (72/9). In Pakistan's provinces of ICT and KP, fewer than 20 percent of teachers are able to do so. Jordan is the only system^{xxi} where more than 80 percent of the teachers can identify the correct answer to $\sqrt{36} - \sqrt{9}$. In only two out of 17 systems,^{xxii} can more than 80 percent of language teachers identify the correct answer to one of the reading comprehension questions, and in only one (Jordan) out of 18 systems can more than 80 percent of teachers correctly identify the appropriate conjunction "unless" in the sentence "*Unless* you tidy up your room, you won't get candy." These are not advanced tasks. Furthermore, they are drawn from content that teachers currently teach or are expected to teach as part of the Grade 4 curriculum.

Across most education systems, at least half of the variation in teacher content knowledge occurs among teachers within the same school. Colleagues teaching in the same school building often have vastly different levels of subject mastery. This pattern is dominant across most surveyed systems (Figure 2.4b), and has direct implications for how systems should respond. In contexts where the gap is primarily between schools, the priority is raising quality in the lowest performing schools. But in systems where the gap is primarily within schools, as it is in most systems, targeted professional development through tailored school-based coaching and mentoring becomes essential. This also implies that in many cases,

^{xxi} Question was not administered in Madagascar.

^{xxii} Question was not administered in Madagascar.

Figure 2.4b: Within- and between-school variation in teacher content knowledge



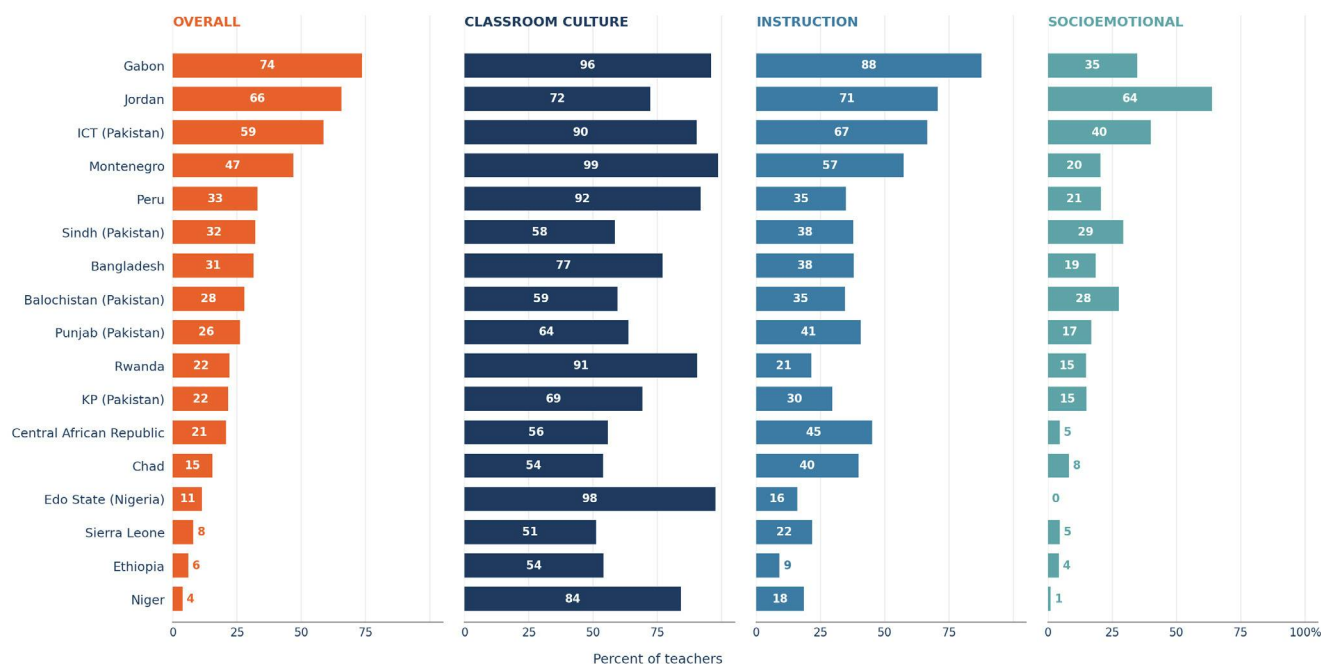
Notes: For each country, a mixed-effects model with a random intercept at the school level is estimated. Teachers are clustered within schools, and no additional covariates are included in the model. Sampling weights are applied at both the teacher and school levels to account for the survey sampling design.

these schools don't have to start from scratch. They already have better performing teachers within their own ranks, which means the expertise to support struggling colleagues already exists inside the school. The challenge is how best to create support structures to unlock it. Initiatives such as Kenya's School-Based Teacher Support System offer a model for fostering this kind of peer-based learning culture, and turning within-school variation from a liability into a resource.³⁷

Effective teaching is not limited to content knowledge proficiency. Teachers must also be able to employ effective pedagogical practices that foster student engagement and support learning.³⁸ The GEPD assesses pedagogical practices in randomly selected Grade 4 classrooms through TEACH Primary,³⁹ a classroom observation tool that captures the quality of teaching practices across three dimensions: classroom culture, instruction, and socio-emotional skill development (Figure 2.5). In most education systems (14 of 17), the majority of teachers do not have satisfactory pedagogical skills.^{xxiii} However, there is wide variation across systems: Gabon, Jordan, and Pakistan's ICT show relatively better performance, while in Ethiopia, Niger, and Sierra Leone, fewer than one in 10 teachers demonstrate satisfactory pedagogical practices.

^{xxiii} Satisfactory practices are defined based on a score of at least 3 on each corresponding item (on a scale of 1–5), as measured by the TEACH Primary classroom observation tool.

Figure 2.5: Percent of teachers with satisfactory pedagogical skills (measured by TEACH Primary)



Notes: Satisfactory practices are defined based on a score of at least 3 on each corresponding item, on a scale of 1–5. Madagascar is excluded due to data unavailability.

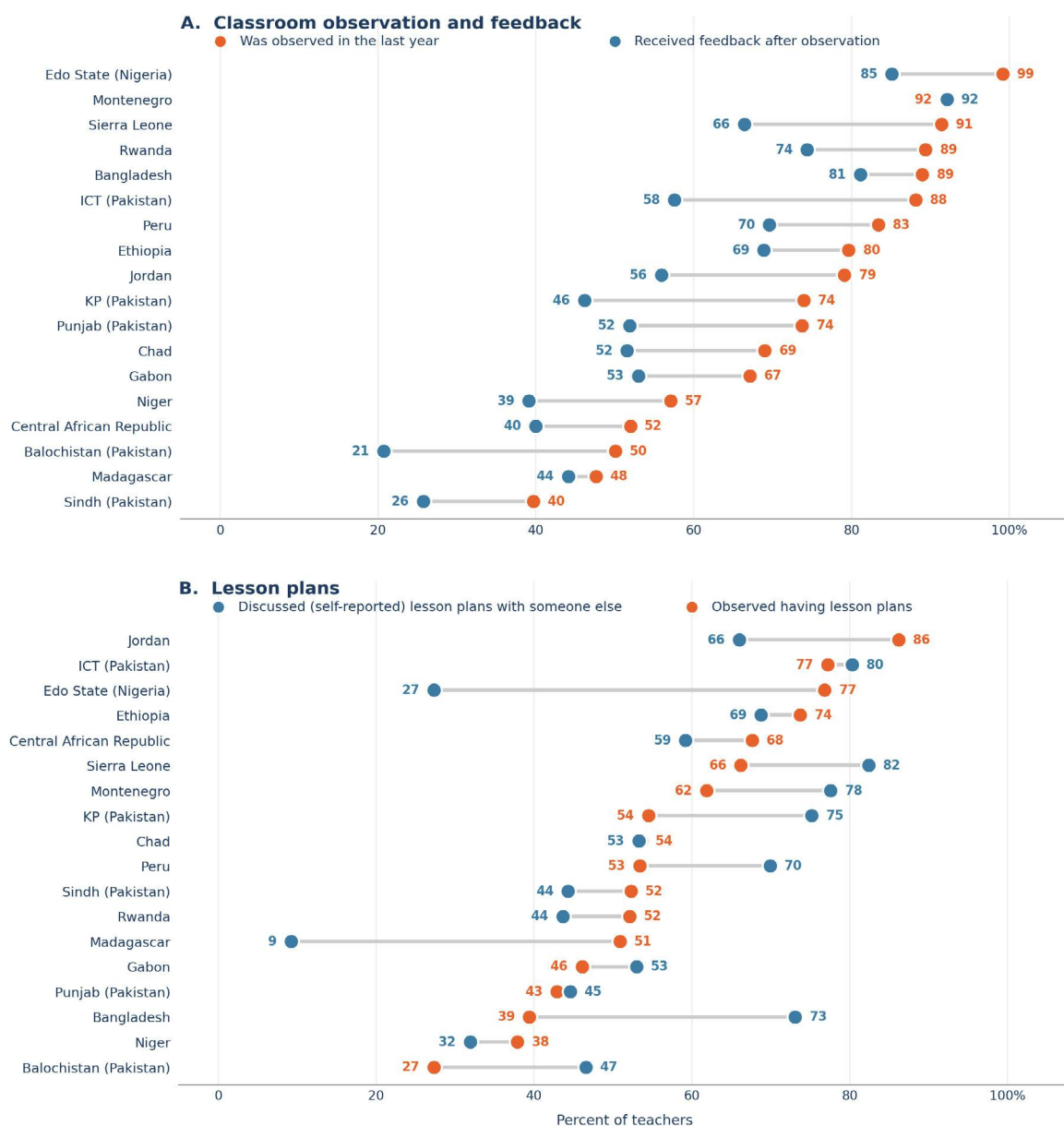
Classroom culture practices are stronger relative to other pedagogical dimensions. Across all systems, except Sierra Leone, at least 80 percent of teachers are observed as treating all students respectfully, and scores on creating a supportive learning environment and setting positive behavioral expectations are consistently higher than other dimensions. But scores drop sharply for instructional practices and fall further for practices aimed at fostering socio-emotional skills. In only two systems (Gabon and Jordan) are more than 50 percent of teachers observed to satisfactorily promote collaboration among students through peer interaction, and in six systems results are in single digits. This aligns with the systematically lower socio-emotional scores on the Grade 1 assessment, suggesting that systems are struggling to develop these skills in children at every stage, in part because teachers themselves lack the pedagogical tools to foster these critical skills.

Key elements of effective teacher support are largely missing from most education systems surveyed. Some of the most effective interventions to improve learning in low- and middle-income countries (LICs and MICs) involve providing teachers with structured lesson plans, aligned learning materials, and ongoing support (Figure 2.6).^{xxiv} Yet across all systems, except Jordan, less than 80 percent of the teachers are observed as having a lesson plan, with availability ranging from 27 percent in Balochistan to 86 percent in Jordan.^{xxiv} Classroom observations are another important aspect of teacher support. In around half the systems (eight out of 18), at least 80 percent of all teachers report that their classroom was observed over the past 12 months, ranging from 40 percent in Sindh to 99 percent in Edo State. But observation without feedback is of limited value. Only in three systems (Bangladesh, Edo State, and Montenegro), do at least 80 percent of teachers report receiving feedback following an observation. In contrast, in Paki-

^{xxiv} Lesson plans must also contain an evidence-aligned sequence of skills and subskills and be aligned with student textbooks to be effective.

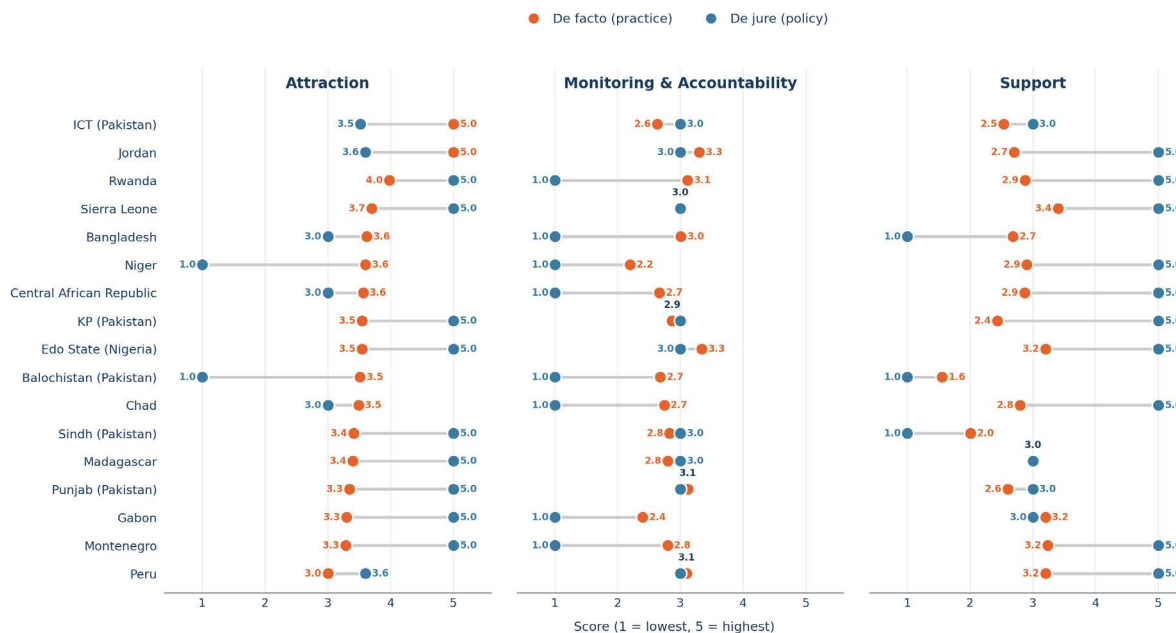
stan's Balochistan and Sindh provinces, only about one in four teachers report receiving such feedback. But feedback alone is not sufficient. It needs to be substantive for it to be effective. While in Montenegro over 90 percent of all teachers report engaging in substantive discussions lasting 30 minutes or more, this proportion does not exceed 10 percent across half the systems surveyed.^{xxv}

Figure 2.6: Teachers supported through lesson plans, classroom observations, and feedback



^{xxv} These statistics on feedback include all teachers (only some of whom were observed).

Figure 2.7: Gap between *de jure* and *de facto* policies impacting teacher effectiveness



Notes: For each policy indicator, there are two numbers reported: one based on *de facto* data, which primarily comes from the School Survey, and one based on the *de jure* data collected through the Policy Survey. Further details are available in the [GEPD Technical Note](#).⁴¹ Estimates for Ethiopia are not reported due to data unavailability for *de jure* indicators.

Policies and regulations related to teacher support often exist on paper, but the mechanisms to implement these policies need to be strengthened. In more than half of surveyed systems (10 out of 17), policies on teacher support are strong on paper but weakly implemented in practice (Figure 2.7). In 12 out of 17 systems, a practicum is required as part of preservice training. Yet only 53 percent of teachers in these 12 systems report completing one. Most systems have policies for in-service training,⁴² yet participation rates range from four percent in Pakistan’s Balochistan province to 72 percent in Peru.

On the other hand, policies on teacher accountability are weaker on paper across many systems, leading schools to develop their own mechanisms. Eight out of 17 systems do not report a policy for collecting data on teacher absenteeism. Yet 86 percent of teachers report that there are consequences for being absent 30 percent of the time, regardless of whether a formal policy exists. Principals and teachers appear to have crafted context-specific, school-level accountability practices that operate independently of the formal policy framework. This suggests that school-level agency can partially compensate for policy gaps, and that accountability mechanisms can be shaped by school leaders.

Finally, teachers often have to contend with challenging classroom environments. The percentage of classrooms with more than 40 students is as high as 56 percent in Rwanda and 37 percent in Chad. Some classrooms in Chad and Pakistan’s Sindh province have more than 130 students, highlighting the difficult environments faced by teachers.^{43,44} Delivering effective instruction under these conditions, let alone implementing differentiated instruction, is difficult. Longer-term solutions to improve student-teacher ratios are essential, but in the immediate term, equipping teachers with pedagogical strategies for managing large classes can make a substantial difference.

Unequal Access: Many Schools Lack Essential Learning Materials and Infrastructure

In environments that lack basic resources, even the best-prepared students struggle to learn, and even the most capable teachers struggle to teach. Functional classrooms, safe drinking water, electricity, and core learning materials like textbooks are prerequisites for effective instruction that leads to learning.⁴⁴ Over the past few decades, many countries have invested heavily in school infrastructure as part of broader efforts to expand access to education, and progress is visible. Basic classroom essentials like functioning blackboards and furniture are found in over 90 percent of classrooms in most systems. But beyond these basics, significant gaps remain in inputs and infrastructure critical for learning (Figures 2.8 and 2.9).

Students need core learning materials to engage with the subject matter. Textbooks, pens, and exercise books give students the opportunity to access content and practice skills. Yet, in the majority of the systems surveyed (11 out of 18), less than 80 percent of Grade 4 students have textbooks, as observed by enumerators. The availability of textbooks also varies considerably across countries, from more than 90 percent of students with textbooks in Bangladesh, Jordan, Montenegro, and in the Pakistani provinces of ICT, KP, and Punjab, to less than 20 percent in the Central African Republic, Madagascar, and Niger (Figure 2.8). In contexts where most students lack textbooks, the immediate priority is procurement and distribution to ensure books reach classrooms. In the longer term, it is vital to build a robust book supply chain, covering planning and forecasting, content development, publishing, printing, and distribution.

Figure 2.8: Availability of core teaching and learning inputs



Notes: The first column shows the proportion of students with textbooks, the second and third columns show the proportion of classrooms equipped with the corresponding items, while the fourth column shows the proportion of schools with access to at least one functional computer. Madagascar is excluded from estimates of basic classroom furniture due to missing data.

Access to basic infrastructure varies widely across and within systems. Ensuring access to basic infrastructure is important because it is a foundational prerequisite for critical aspects of the learning experience, from teacher and student engagement to digital learning. Yet in 11 out of 18 systems, fewer than 80 percent of schools have access to functional electricity. The variation across countries is large: in the

Central African Republic, Chad, Niger, and Sierra Leone, fewer than 10 percent of schools had functional electricity at the time of the survey, whereas in Jordan, Montenegro, and Pakistan's ICT and Punjab, coverage is almost universal (above 95 percent). Pronounced urban-rural gaps in electrification are seen in some contexts, such as Gabon (77 percentage points) and Sindh (52 percentage points). But data show that it is possible to make progress on narrowing these gaps. For example, in Ethiopia, the urban-rural electrification gap decreased by approximately 24 percentage points between 2020 and 2025. Initiatives such as the Mission 300, which aims to connect 300 million people to electricity across Africa by 2030, are critical for accelerating school electrification and reducing urban-rural inequity.⁴⁵

Sanitation and water reflect a story of unfinished business. No surveyed system reports functioning toilets^{xxvi} in more than 85 percent of schools, with coverage in single digits in Madagascar, and Pakistan's Balochistan and Sindh provinces. This matters, not least because improved sanitation is associated with increased school participation and attendance, especially for girls. In particular, the availability of sex-specific toilets can improve the enrollment and retention of adolescent girls.⁴⁶ Yet, only three systems (Montenegro, Peru, and Rwanda) report having separate toilets for girls in at least 90 percent of schools. Access to drinking water is relatively better across systems, with more than 70 percent of schools having drinking water across more than half the surveyed systems. However, there is wide variation across systems, ranging from approximately 19 percent in Edo State (Nigeria) to over 95 percent in Pakistan's Punjab province and Montenegro. Integrated water, sanitation, and hygiene (WASH) interventions have demonstrated potential to reduce student absenteeism and dropout.⁴⁷

Figure 2.9: Availability of basic infrastructure



Notes: All columns present the percentage of schools equipped with the corresponding item.

^{xxvi} A school is considered having a functioning toilet if it meets all of the following criteria: (1) it is equipped with at least one of the following sanitation facilities: (i) a flush or pour-flush toilet; (ii) a pit latrine with or without a slab; (iii) a composting toilet; (iv) a hanging latrine; or (v) a bucket latrine; (2) at least some of these facilities are usable and in working condition; (3) separate toilets or latrines are available for boys and girls in co-educational schools; (4) the facilities ensure privacy; and (5) handwashing facilities are available on the school premises.

Internet connectivity is limited. In half the systems surveyed (nine out of 18), less than 10 percent of the schools have access to the internet, with Montenegro being the only system where coverage exceeds 85 percent. As countries take steps to keep pace with rapid advances in education and the adoption of digital technology in the workplace, it is critical to ensure that digital investments do not exacerbate existing inequalities, particularly by further marginalizing rural and underserved populations.

Technology, including adaptive learning software or AI-assisted tutoring, has the potential to transform teaching and learning.⁴⁸ But realizing that potential requires foundational digital infrastructure that many systems lack, such as electricity, the internet, and functional computers. Significant disparities persist in schools' access to functional computers. While more than 90 percent of schools in Edo State (Nigeria), Jordan, Montenegro, and Pakistan's Punjab province have access to functional computers, access is minimal to nonexistent across close to half the systems surveyed (eight out of 18). Disparities also exist within countries. In Ethiopia, the urban–rural gap in access to functional computers is more than 45 percentage points, with wide gaps in Gabon and Peru as well. However, progress is possible: Ethiopia improved the percentage of schools with functional computers by 17 percentage points between 2020 and 2025, largely driven by an increase in rural coverage (from seven to 24 percent).

The gap between the inputs and infrastructure that children need and those they can actually access is seldom the manifestation of a gap in policy. *De jure* policies related to basic inputs and infrastructure, including textbooks, water, and toilets,^{xxvii} exist in all but two systems surveyed. The challenge lies in ensuring that policies on paper translate into resources in classrooms. Where policy frameworks are notably absent, however, is in education technology. Only four out of 17 systems have policies in place governing access to education technology. As the role of technology in teaching and learning grows, establishing policy and regulatory frameworks to guide its use is a necessary first step. But for many systems, the more immediate priority remains ensuring that the basic infrastructure and inputs, such as electricity, sanitation, drinking water, and textbooks, reach every school.

The Leadership Challenge: School Leaders Need Better Information and Support to Lead Effectively

School leaders play a key role in realizing learning.⁴⁹ Effective school management helps align and operationalize critical school-level drivers of learning described earlier, including learner readiness, teacher effectiveness, and access to essential inputs and infrastructure. Research shows a link between management quality and teacher effort, engagement, and student achievement.⁵⁰ While school management may not directly impact student learning outcomes, it influences them indirectly by prioritizing learning goals, supporting teacher instruction, and mobilizing and managing essential school resources.

School management stands out as a key differentiator of top-performing schools (Figure 2.10). Operational management, which captures the presence and quality of core operational management functions, shows some of the largest differences between the highest and lowest performing schools, reaching over 25 percentage points. The knowledge and management skills of principals also show differences of over 15 percentage points. Other critical factors with substantial gaps between the highest and lowest performing schools, such as the availability of essential inputs, infrastructure, and teacher pedagogical practices (measured by the TEACH Primary score), are commonly shaped by effective school leadership. As instructional leaders, principals play a key role in mobilizing essential resources and providing instructional support to teachers.

^{xxvii}*De jure* policies related to functioning toilets are only missing in one of the surveyed systems.

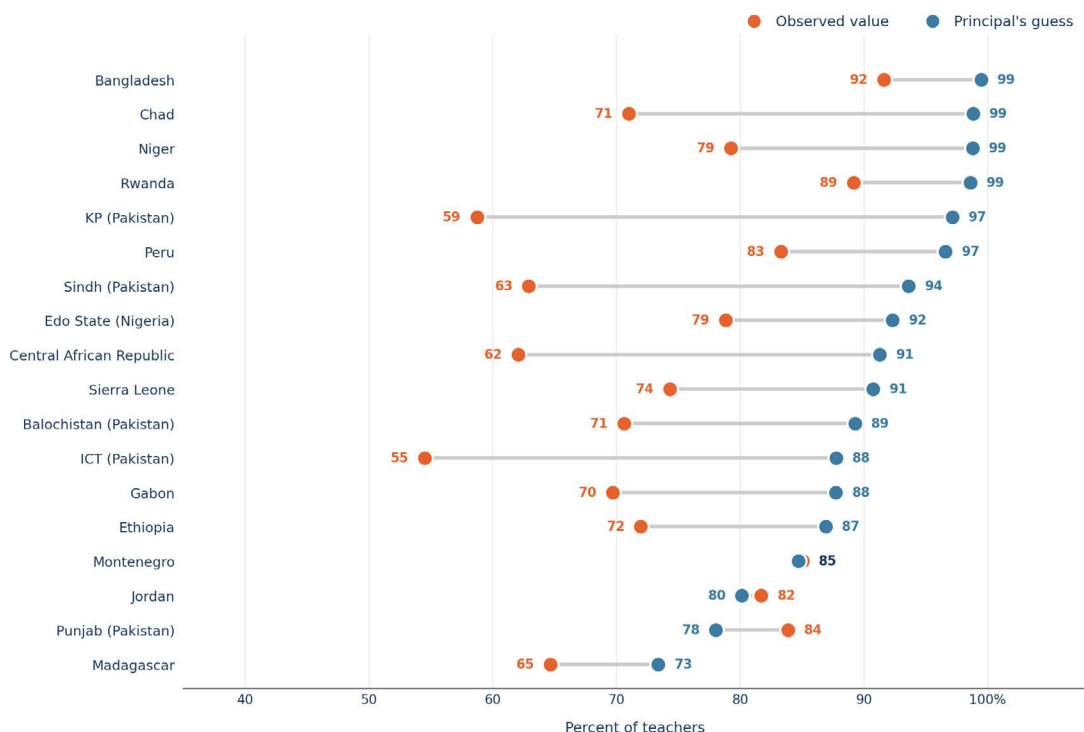
Figure 2.10: Differences in selected core indicators between the highest and lowest performing schools



Notes: We define a school as low-performing if its aggregated student performance falls within the bottom 20 percent of the cross-country distribution, and as high-performing if it falls within the top 20 percent. For each indicator, we calculate the percentage of low- and high-performing schools that score at least 4 for the corresponding indicator. In the figure, the percentage of low-performing schools scoring at least 4 are represented by blue circles, and high-performing schools scoring at least 4 represented by gray squares. Only indicators on a 1–5 scale are included in this graph.

Yet principals often lack an accurate picture of the schools they lead. In all but three systems surveyed (Jordan, Montenegro, and Pakistan’s Punjab province), principals overestimate the percentage of teachers in their schools who can solve simple numeracy and literacy questions, with gaps as large as 30 percentage points in Pakistan’s provinces of ICT, KP, and Sindh (Figure 2.11). In KP, principals report near universal teacher proficiency on select questions from the basic literacy and numeracy assessment completed by teachers, yet actual results reveal that only around 60 percent of teachers demonstrate this proficiency in practice. This mirrors the perception gaps documented among public officials in the next chapter, where officials overestimate class sizes and underestimate teacher absenteeism. At every level of the education system, from ministry headquarters to the principal’s office, decision-makers are operating with a distorted picture of frontline realities.

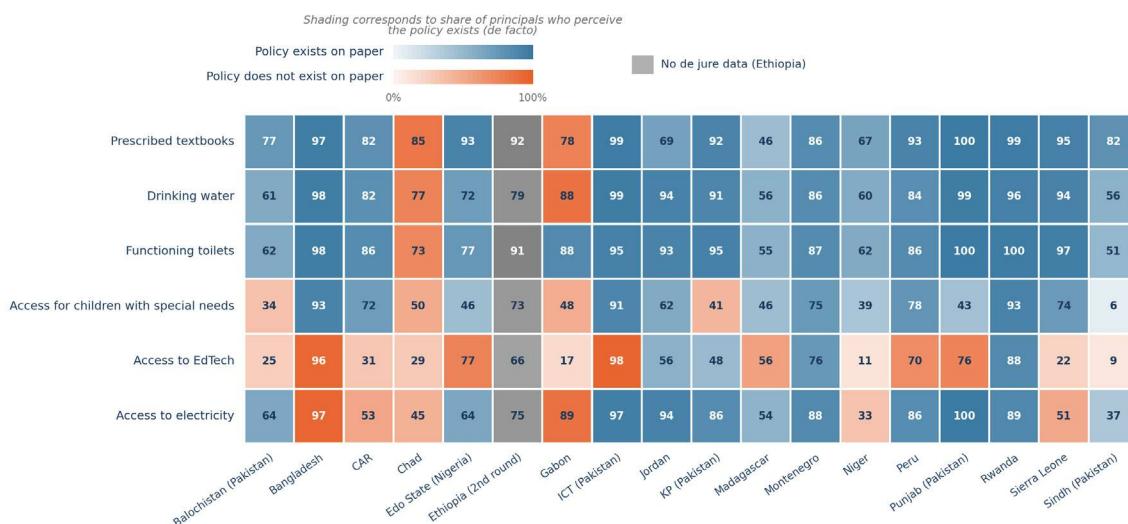
Figure 2.11: Principals' perception of teachers' content knowledge



Notes: Indicators are derived from school-level data and aggregated to the country level using sampling weights. "Observed value" refers to the average percentage of teachers who correctly answer the sample set of content knowledge questions. "Principal's guess" refers to principals' school-level estimates of the same metric.

These perception gaps are not only limited to teachers' content knowledge but also extend to principals' awareness of the policies governing critical inputs and infrastructure. As mentioned above, *de jure* policies related to basic inputs and infrastructure, including textbooks, water, and toilets, exist in all but two systems surveyed. But principals' knowledge of these policies vary considerably across countries (Figure 2.12). This matters because principals are uniquely positioned to translate policy into practice at the school level. For example, principals who are unaware that a textbook provision policy exists are far less likely to advocate for its implementation, increasing the risk that students remain without essential learning materials.

Figure 2.12: Implementation gap in policies governing inputs and infrastructure and principals' perceptions of the existence of such policies

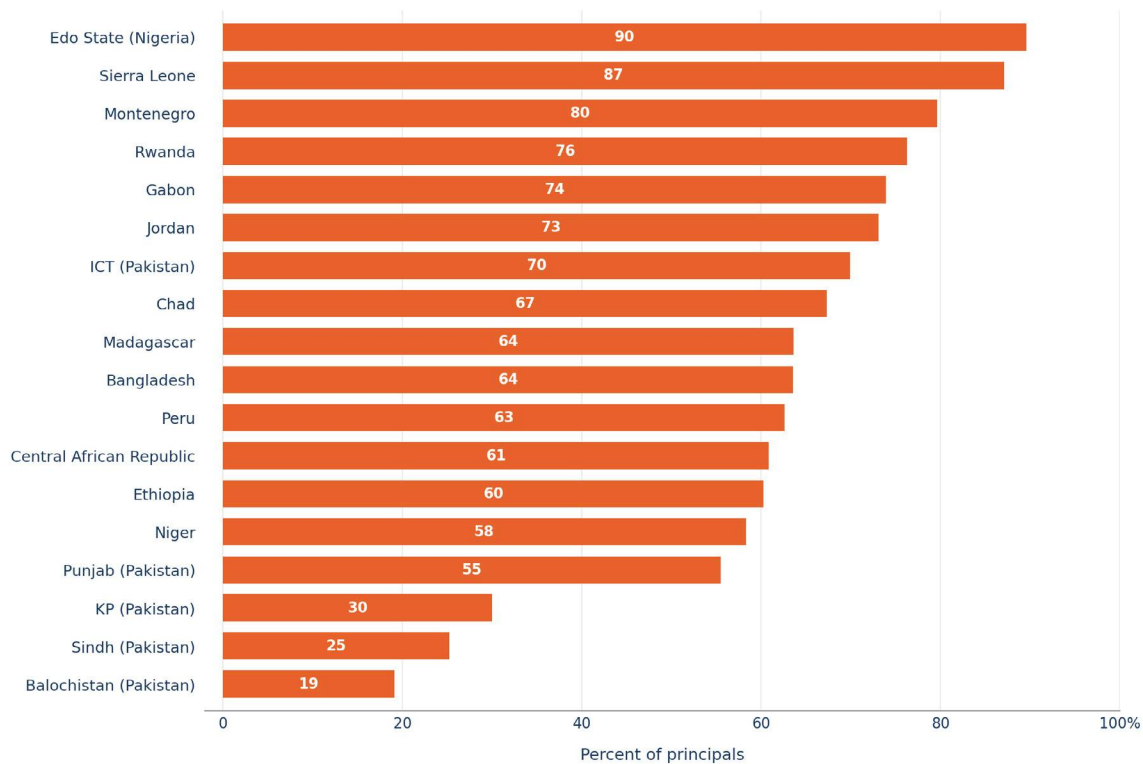


Note: The color of each cell represents whether a given policy requiring the availability of a specific input/infrastructure item exists on paper. Cells are color-coded to reflect de jure policy status: blue cells denote policies that exist on paper, and orange cells denote policies that do not exist on paper. The shading of each cell corresponds to the share of principals who perceive that the policy exists (shown as a percentage within each cell). De jure data for Ethiopia (2025) is not available.

When principals lack key knowledge, efforts to improve learning can stall. Across most systems (16 out of 18), more than 70 percent of principals report having school-level goals focused on improving student scores or exam pass rates. Yet, in 14 out of 18 systems, less than 30 percent of principals report that their schools have goals explicitly aimed at supporting teachers to improve pedagogical practices, which is a critical condition for sustained progress in learning outcomes. Similarly, principals play a key role in ensuring the availability of essential school resources. Yet, in four out of five of the systems with the lowest textbook availability (below 50 percent), fewer than half of the principals report taking action to ensure textbook access for students who lacked them during the first month of the school year.

Supporting principals in effectively managing their schools can be a cost-effective strategy to improve student learning and teacher performance,⁵¹ yet far too many systems are failing to do so. Research shows that targeted training programs can improve both the performance of the principal and student learning outcomes.⁵² Yet many systems have yet to use this lever effectively to strengthen school leadership. While over 60 percent of principals report receiving training in school management in most systems, coverage is lower in all provinces of Pakistan except ICT, falling to just 19 percent in Balochistan. In contrast, Sierra Leone and Edo State (Nigeria) report coverage above 80 percent (Figure 2.13). Access to mentoring or coaching from experienced principals is also scarce. No education system surveyed offers such opportunities to more than half of principals, and even systems that generally perform better in other aspects, such as Peru and Montenegro, show low coverage in this domain (under 17 percent). These findings highlight opportunities for expanding training and mentorship opportunities for principals.

Figure 2.13: Share of principals reporting prior training in school management



3

Inside the Education Bureaucracy— Views from 3,500 Public Officials



Education systems extend far beyond classroom walls. The ministries, departments, and offices that set policies, allocate resources, manage personnel, and monitor performance shape how effectively reforms translate into learning outcomes. This administrative layer is massive. Globally, approximately 75 percent of education personnel work in the public sector. In many countries, the health and education sectors often exceed 50 percent of total government employment. These sectors also consume substantial shares of national budgets,⁵³ with education's share alone averaging roughly four percent of GDP.⁵⁴ When bureaucracies function poorly, the consequences reverberate through systems and sectors, from large-scale resource misallocation to reduced effectiveness of teachers and principals. The quality of bureaucratic machinery can shape whether ambitious education reforms succeed, whether resources reach classrooms with the greatest need, and ultimately whether children learn.

The GEPD survey of public officials offers a rare window into education bureaucracies. Public officials are typically the consumers of data, as they review school assessments, analyze attendance figures, and scrutinize budget reports, but they are rarely the subjects of systematic inquiry. The GEPD's Survey of Public Officials addresses this evidence gap through the collection of responses from over 3,500 education officials across 19 education systems (Table 3.1) and multiple government levels, from ministry headquarters to district offices. Field response was overwhelmingly positive. Officials welcomed the chance to voice concerns, share perspectives, and articulate the operational challenges they face in their day-to-day roles. Table 3.1 reveals a highly educated administrative workforce, with 59 percent of officials holding a master's degree or higher. Officials report an average of 20 years in civil service and roughly 10 years in their current organizations, reflecting limited organizational mobility within systems. The predominantly male dominant composition of this workforce (75 percent across surveyed systems, and exceeding 80 percent in Chad, Ethiopia and Sierra Leone) highlights persistent gender imbalances in administrative roles that shape education policy and implementation.

Table 3.1: Demographic characteristics of interviewed officials

	Age	Years in current position	Years in current organization	Years in civil service	Hold master's degree or higher (%)	Male (%)
Bangladesh	50	4	7	21	99	79
Central African Republic	49	4	11	18	46	86
Chad	49	4	8	19	22	90
Edo State (Nigeria)	44	3	11	17	18	56
Ethiopia	42	5	10	19	56	87
Gabon	50	4	11	22	26	74
Jordan	48	4	17	21	73	72
KP (Pakistan)	46	2	11	20	92	71
Madagascar	45	4	8	16	58	71
Montenegro	50	9	13	22	27	19
Mozambique	48	4	10	24	22	73
Niger	52	5	8	25	38	81
Balochistan (Pakistan)	46	1	10	22	97	78
ICT (Pakistan)	47	4	11	19	100	79
Punjab (Pakistan)	50	2	13	24	100	68
Sindh (Pakistan)	53	2	26	30	98	84
Peru	47	3	7	19	62	56
Rwanda	42	3	6	13	40	75
Sierra Leone	45	3	4	10	42	82
Overall	48	4	10	20	59	75

Notes: "Overall" represents a simple average across education systems.

The path from policy to practice is not linear. Policies designed at the central Ministry must cascade through multiple administrative levels, to be interpreted by thousands of officials—with widely varying skills and resources at their disposal—before ultimately shaping the daily decisions of principals and teachers in diverse classrooms. Weak links along the implementation chain can undermine even well-designed reforms. Understanding the administrative realities facing education officials helps identify what is not working and where systems can improve.

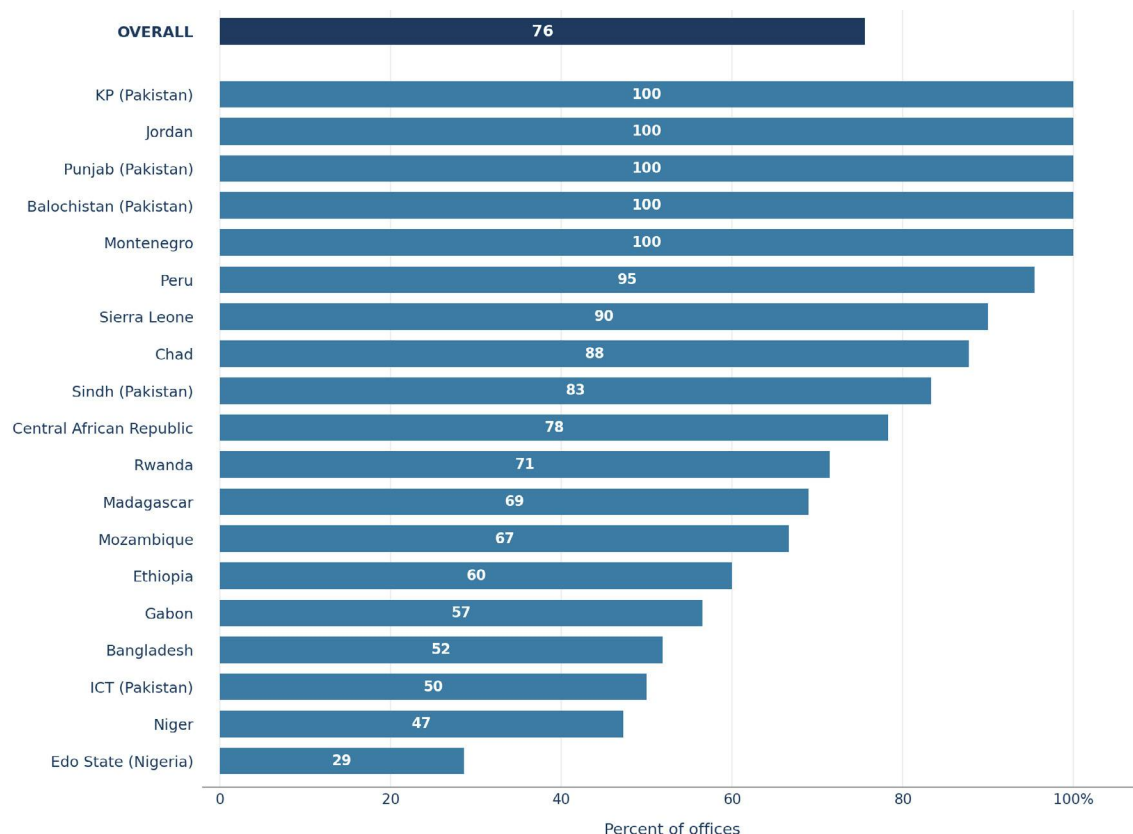
Leadership Instability: High Turnover Erodes Institutional Memory and Reform Momentum

Frequent changes in leadership plague education systems. Ministerial turnover disrupts strategic planning and forces bureaucracies into repeated cycles of reorientation as new leaders arrive and shift priorities.⁵⁵ Peru exemplifies the most extreme case among surveyed systems. The country cycled through 12 different education ministers between 2019 and 2025, with an average tenure under six months.⁵⁶ The consequences are predictable. Institutional memory struggles to take root, long-term reforms lose momentum, and officials often spend more time adapting to new directives than implementing existing ones. For example, in South Africa, high levels of leadership and administrative turnover have undermined efforts to professionalize public services, placing additional strain on political and administrative functioning.⁵⁷

There is high leadership turnover across education bureaucracies, with Peru cycling through twelve education ministers in six years.

Instability is not limited to the top. Leadership turnover is common across every level of administration. Across the systems surveyed, more than half of public officials report that their head of department had changed within the previous year (Figure 3.1). More than three in 10 officials (32 percent) saw a new person take on the role of finance director, and nearly one in three (30 percent) reported a new planning director during the same period. These frequent shifts across key leadership and operational positions add to the disruption caused by ministerial turnover. Systems lose not only strategic continuity but also the technical expertise and institutional knowledge that can take years to build.

Figure 3.1: Director-level turnover in the 12 months preceding interview



Notes: "Overall" represents a simple average across education systems. Director-level position turnover is defined as a change within the past year in one or more of the following positions: (i) head of the education ministry or department, (ii) finance director, (iii) planning director, (iv) school supervision director, or (v) M&E director.

Unreliable electricity, scarce computers, and spotty internet make officials' jobs harder, and in some cases nearly impossible. On average, officials report having electricity during three-quarters of working hours (74 percent). The range is vast. Montenegro (100 percent) and Peru (95 percent) report near-universal coverage, while the Central African Republic (18 percent) and Nigeria's Edo State (21 percent) face severe shortages. Internet connectivity sufficient for email services functions is available for only 48 percent of working days across all surveyed systems, as reported by officials. Barely more than half of officials (56 percent) have access to a computer (Table 3.2). These infrastructure gaps matter profoundly for motivation and performance.⁵⁸ Research consistently links poor working conditions to lower productivity, more employees considering leaving their jobs, and weaker organizational performance.⁵⁹

Table 3.2: Access to basic office infrastructure, as reported by officials

	Working hours with electricity (%)	Working days with working phone network (%)	Working days with functioning internet (%)	Officials with computer access (%)	Officials with access to a vehicle (%)
Bangladesh	79	92	56	55	41
Central African Republic	18	83	24	27	8
Chad	34	35	12	23	8
Ethiopia	66	88	40	58	18
Gabon	88	73	46	45	14
Jordan	91	96	59	84	55
Madagascar	88	97	53	41	12
Montenegro	100	100	50	100	10
Mozambique	94	66	37	56	41
Niger	79	80	47	44	32
Edo State (Nigeria)	21	82	52	15	34
Balochistan (Pak.)	70	90	54	67	61
ICT (Pakistan)	69	100	63	87	86
KP (Pakistan)	73	100	57	97	84
Punjab (Pakistan)	93	98	58	90	68
Sindh (Pakistan)	75	90	40	60	30
Peru	95	84	57	100	41
Rwanda	85	85	54	99	61
Sierra Leone	85	90	53	72	76
Overall	74	84	48	56	37

Notes: “Overall” represents a simple average across education systems. Electricity is defined as power supplied by grid or generator; a functioning phone network is defined as network connectivity that enables a call success rate of 50 percent or higher; computer access is defined as access to a desktop or laptop computer; functioning internet is defined as internet access sufficient to access email services; and vehicle access refers to a vehicle (privately owned or institutional vehicle) that can be used for work purposes.

Three out of four officials believe they’d earn more in the private sector. Yet only one in four want to leave.

Technical skills often fail to match formal qualifications. While 59 percent of officials hold a master’s degree or higher, proficiency in basic administrative tasks remains low. According to human resource (HR) managers in surveyed offices, only 59 percent of officials can use a computer to draft a memo, 45 percent can prepare a PowerPoint presentation, and 48 percent can produce an Excel spreadsheet (Table 3.3). The variation across systems is stark. In Peru and Rwanda, HR managers report that around 90 percent of officials are proficient in these tasks. In contrast, in Chad and Pakistan’s Sindh province, HR managers report that fewer than one in three officials can perform them. These gaps in digital proficiency suggest that officials are not receiving adequate training and support for data-driven decision-making and policy implementation.

Despite these challenging conditions, public officials demonstrate remarkable commitment. Four out of five officials believe they would earn more in the private sector. Yet, when asked whether they want to move into the private sector within the next two years, only one in four want to leave. The variation is telling. In Bangladesh, Madagascar, and Pakistan’s Punjab province, fewer than one in ten officials express.

Table 3.3: Proportion of public officials reporting proficiency in basic tasks

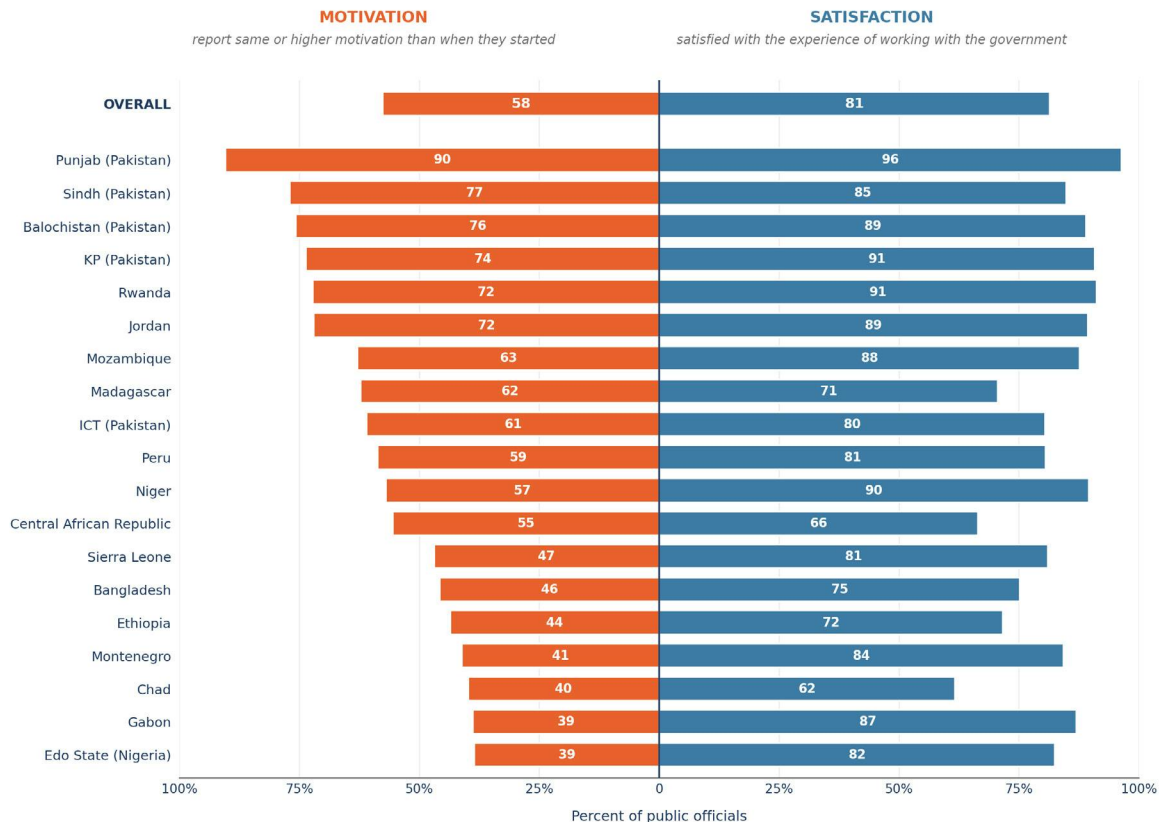
	Officials able to use a computer to write a memo (%)	Officials able to make a Power Point presentation (%)	Officials able to make an Excel spreadsheet (%)
Bangladesh	57	48	45
Central African Republic	36	22	26
Chad	30	17	19
Ethiopia	73	56	60
Gabon	57	27	34
Jordan	73	57	74
Madagascar	62	26	58
Montenegro	100	30	20
Mozambique	73	41	39
Niger	48	26	39
Edo State (Nigeria)	51	21	33
Balochistan (Pakistan)	45	48	45
ICT (Pakistan)	83	63	65
KP (Pakistan)	83	66	33
Punjab (Pakistan)	71	63	49
Sindh (Pakistan)	30	18	23
Peru	92	91	91
Rwanda	99	86	96
Sierra Leone	77	36	38
Overall	59	45	48

Notes: "Overall" represents a simple average across education systems.

a preference for the private sector. Across fragile and conflict-affected settings, the figures are higher, reaching approximately 45 percent in Ethiopia and Niger. When asked what might lead them to exit the civil service, 56 percent cite wages, 38 percent point to conditions of service, 24 percent mention limited career opportunities, 22 percent point to poor working environments, and 20 percent highlight inadequate capacity development. The fact that so many officials stay in the public sector despite lower pay reflects the appeal of job security, other nonmonetary benefits, and generous pensions—factors that become especially important in unstable economic environments.

Fifty-eight percent of officials indicate that their motivation has remained steady or grown since taking on their current roles, while 81 percent report being satisfied with their work in the public sector (Figure 3.2). Trust among colleagues is high, with 76 percent reporting mutual trust in fulfilling their commitments, a key enabler of organizational effectiveness. More than half of all officials also report that their offices encourage innovation and have experience adopting new administrative practices, suggesting that a culture of problem-solving and adaptation is emerging, even in low-resource and high-turnover environments. This commitment is an asset most education systems have barely tapped. Addressing basic infrastructure gaps, providing adequate professional development opportunities, and introducing career development frameworks with clear promotion pathways could unlock greater productivity from a workforce that has already demonstrated its dedication to public service.

Figure 3.2: Level of motivation and job satisfaction among public officials



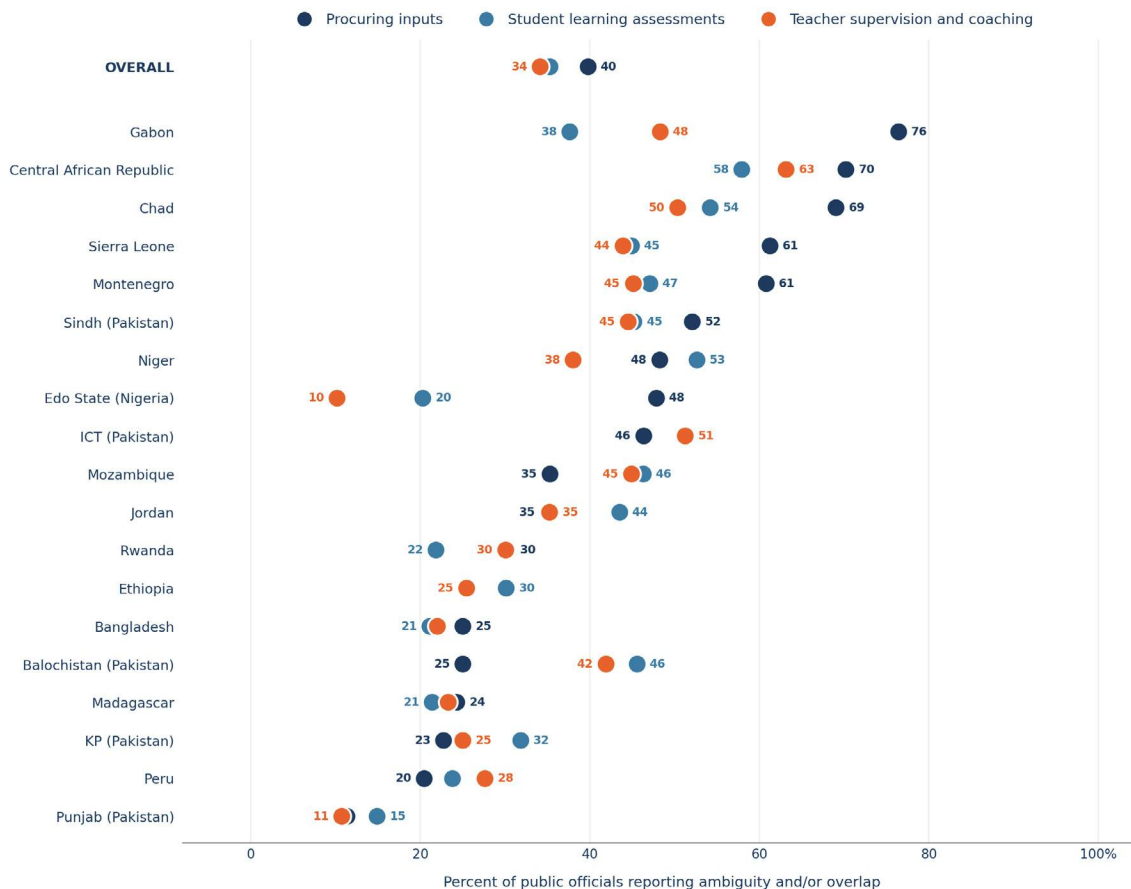
Notes: “Overall” represents a simple average across education systems. Based on the responses from 3,053 and 3,188 officials for the motivation and satisfaction data respectively (513 and 378 officials are excluded because they did not comment on motivation satisfaction respectively).

Merit Under Pressure: Political Interference and Weak Incentives Undermine Personnel Decisions

Confusion over responsibilities undermines coordination and blurs accountability. Effective management depends on clear mandates that define the allocation of responsibilities across administrative levels.⁶⁰ Yet, across many of the surveyed education systems, officials report persistent ambiguity in institutional roles and responsibilities, limited shared understanding of functional boundaries, and uneven clarity on core responsibilities across levels of government.

Public officials frequently report that unclear key education responsibilities overlap or spread across multiple departments and levels of government. Forty percent of officials report ambiguity and overlap in responsibility for procuring inputs, followed by 35 percent in student assessment functions and 34 percent in teacher supervision and coaching. (Figure 3.3). The problem stems from multiple sources, including gaps in legislation that do not clearly assign institutional mandates and uneven awareness of formal roles and responsibilities among officials themselves. Research across four Latin American countries found that officials at national, subnational, and local levels failed to correctly identify 10 to 80 percent of legally assigned tasks, while simultaneously reporting ownership of 15 to 35 percent of tasks allocated to other administrative levels.⁶¹ Functional ambiguity can lead to both inaction and duplication.

Figure 3.3: Percentage of officials reporting unclear or overlapping key responsibilities

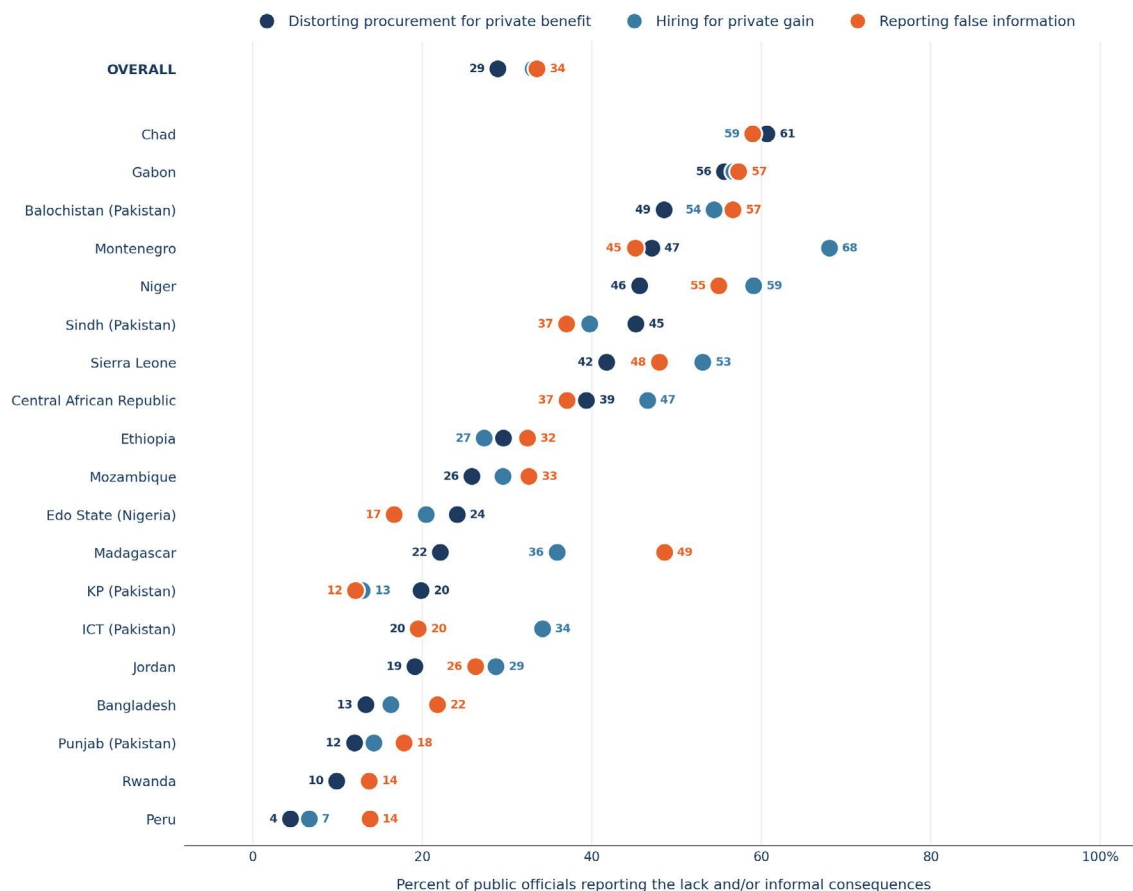


Notes: "Overall" represents a simple average across education systems. Based on responses from 3,189, 3,200, and 3,199 officials regarding procuring inputs, student learning assessments, and teacher supervision and coaching respectively. 377, 366, and 367 officials are excluded due to missing or refused responses for each category. For each category, "ambiguity and overlap" includes the following responses: "responsibility not clear," "responsibility not always clear," "responsibility is clear but there is an overlap in functions," and "do not know."

For example, when a district official expects the ministry to lead teacher recruitment while the ministry expects districts to manage it, vacancies may remain unfilled, resulting in staffing shortages at the school level. Similarly, overlapping mandates can generate duplicated administrative effort, where multiple offices engage in the same task without coordination, reducing efficiency and diluting accountability.

Accountability systems often lack consequences. When responsibilities are clearly defined, officials must face meaningful consequences for failing to fulfill them. Yet across the surveyed systems, substantial numbers of public officials perceive that misconduct would go unpunished or trigger only informal or non-systematic repercussions. Twenty-nine percent of officials report there would be no formal consequence for distorting procurement processes for private gain. Thirty-three percent report the same for hiring teachers based on favoritism rather than merit, and an equal share perceive that reporting false information would not result in formal sanctions (Figure 3.4). The rewards and sanctions that shape public sector performance are rooted in a country's institutional accountability frameworks.⁶² When officials believe, or worse, know from experience, that misuse of authority carries no formal penalty, incentives to perform effectively or act with integrity are substantially weakened. The absence of meaningful consequences for

Figure 3.4: Percentage of officials who perceive no consequences for misuse of authority

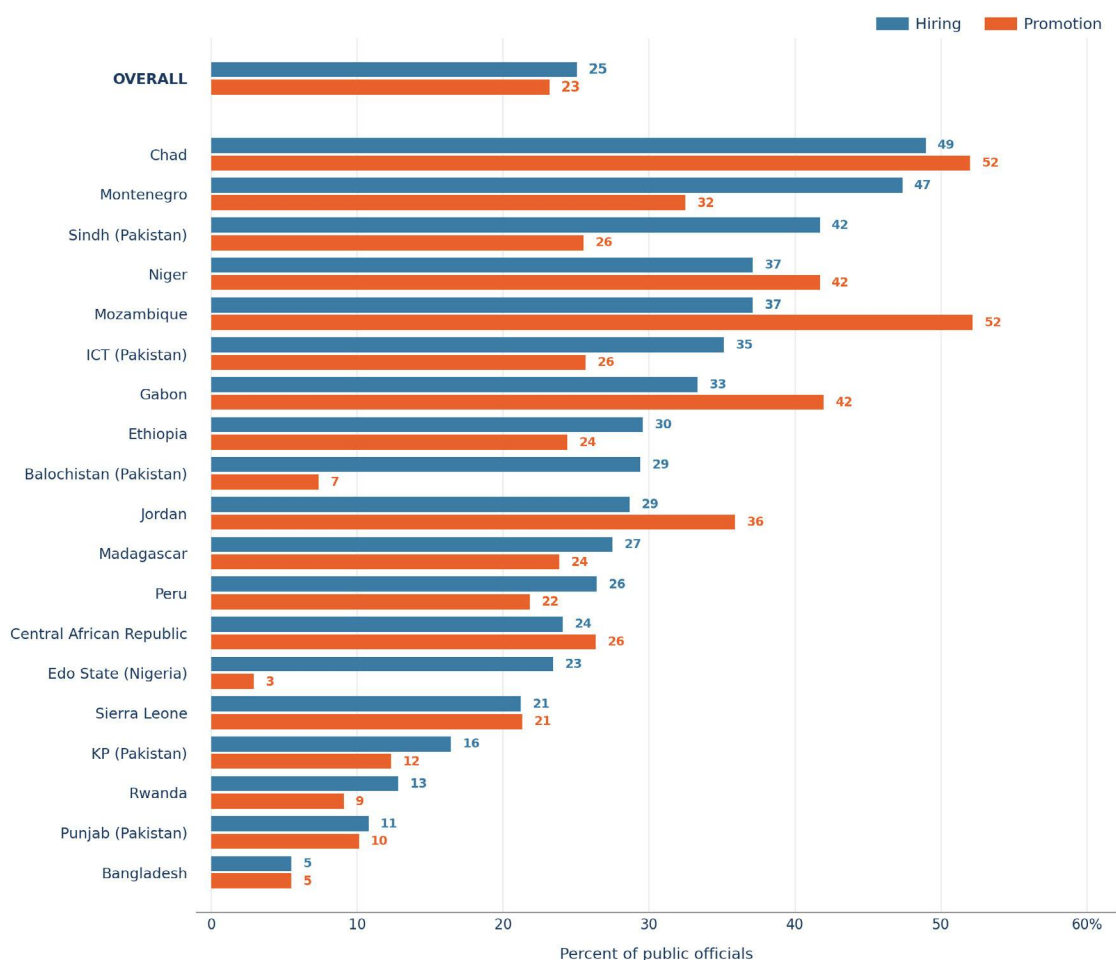


Notes: “Overall” represents a simple average across education systems. Based on responses from 3,197, 3,184, and 3,186 officials regarding hypothetical scenarios on consequences for reporting false information, hiring a teacher for private gain, and distorting procurement for private benefit respectively. 369, 382, and 380 officials are excluded due to missing or refused responses for each category. For each category, “lack or informal consequence” includes the following responses: “no one is held accountable,” “informally held accountable,” “held accountable with unclear consequences,” and “do not know.”

falsifying information is particularly damaging. If officials can manipulate data without formal sanctions, monitoring systems are weakened, and the conditions for evidence-based decision-making become impossible.

Political interference presents another challenge that can distort implementation. The relationship between politicians and bureaucrats is central to how the public sector functions, particularly in the selection and motivation of public employees. Clientelism, the provision of jobs, contracts, and other benefits in exchange for political support, are common features of the public sector across countries in all regions and income groups.⁶³ One quarter of officials (25 percent) report that hiring decisions favor political power over merit, and 23 percent report the same for promotion decisions (Figure 3.5). This politicization has profound consequences. Political interference in personnel management can disrupt the effective functioning of the administration and its schools,⁶⁴ while also potentially reducing the motivation and satisfaction of civil servants affected by clientelistic practices.⁶⁵ These practices may partly explain the gap between officials’ formal qualifications and practical capabilities noted earlier. When hiring and

Figure 3.5: Percentage of public officials reporting the influence of politics on hiring and promotion



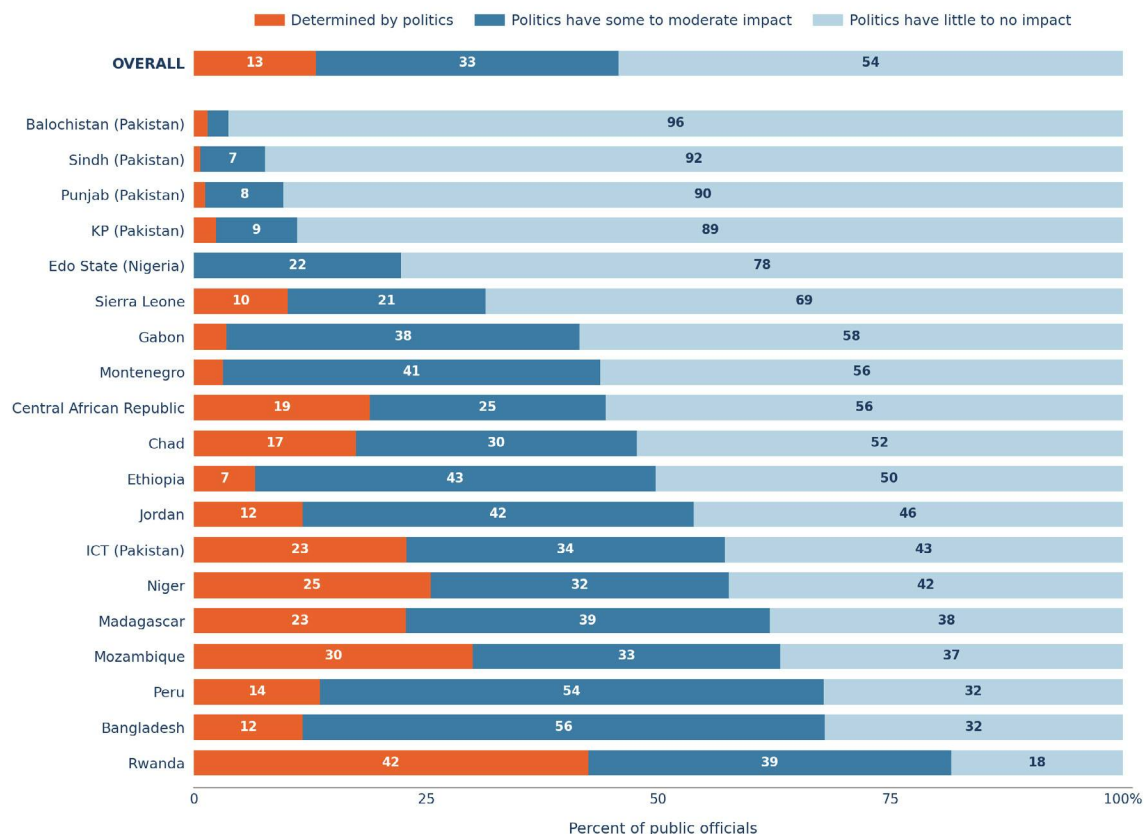
Notes: “Overall” represents a simple average across education systems. Based on the responses from 3,011 and 3,049 officials respectively (557 and 519 are excluded from hiring and promotion statistics respectively for lack of response).

promotion prioritize political connections over demonstrated skills, the workforce’s technical competencies—such as using basic digital tools for data management and communication—may receive less focus.

Political interference is especially pronounced in curriculum design and development. Curriculum is among the most politically charged aspects of education systems, with decisions about what children learn, including history, science, and language of instruction, even triggering conflicts. In recent years, curriculum has been the subject of heated national debates across dozens of countries, including Ghana, Guinea, Pakistan, South Africa, and Uganda. Language of instruction is particularly contentious. Thirty-seven percent of students in low- and middle-income countries are taught in a language they do not understand, and debates about which languages should be used for instruction often become politicized, despite clear evidence that children learn

One in 4 public officials report that hiring and promotion decisions favor political power over merit.

Figure 3.6: Percentage of public officials reporting political influence on curriculum

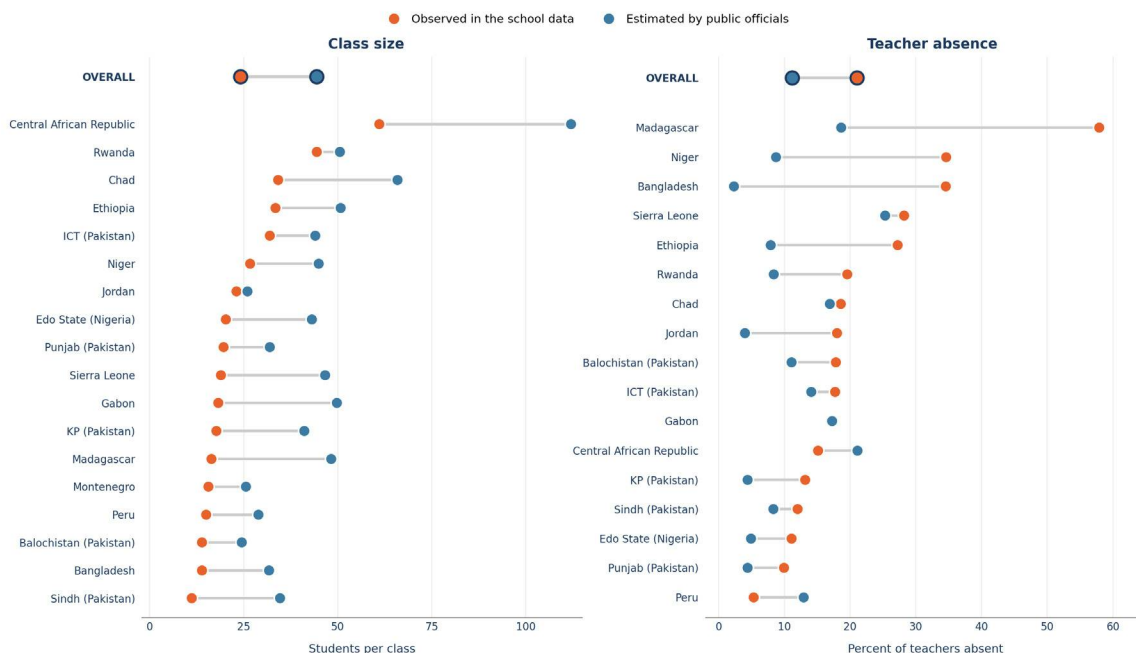


Notes: “Overall” represents a simple average across education systems. Based on responses from 3,020 officials; 546 officials are excluded due to no response.

best when taught in languages they speak and understand.⁶⁶ Across all surveyed education systems, nearly half (46 percent) of public officials report that politics influences curriculum design and development (Figure 3.6). The variation is striking. In Edo State, Nigeria, only 22 percent report political influence, compared to education systems in Bangladesh, Peru, and Rwanda where a majority of officials (exceeding 60 percent) report political influence over curriculum decisions.

Sometimes the challenge is not external pressure but internal perception. A significant gap exists between how officials perceive frontline conditions and what school observations reveal. Public officials estimate that the typical classroom has 44 students, yet school observations reveal the actual figure is 24, a difference of 20 students (Figure 3.7). Similarly, officials estimate teacher absenteeism at 11 percent, while actual rates stand at 22 percent. There is significant variation across systems. In Rwanda and Pakistan’s Balochistan province, officials’ perceptions closely match reality, while in the Central African Republic there is a marked misperception of class size, with the estimated class size being incorrect by more than 50 students.

Figure 3.7: Perceived versus actual class size and teacher absence level



Notes: "Overall" represents a simple average across education systems. The data on observed teacher absence are not reported for Montenegro. Mozambique is excluded due to unavailability of school data. Observed teacher absence values include the data on teachers who are scheduled to be teaching.

Officials' beliefs also shape what happens in classrooms. Research across developing countries shows that beliefs about teaching and learning can directly impact student outcomes and shape policy implementation, yet these beliefs are seldom measured in impact evaluations.⁶⁷ Public officials who normalize teacher absenteeism tend to observe higher absenteeism in practice. Madagascar is a case in point. It reports both the highest proportion of officials who consider it is acceptable for teachers to be absent if they are doing something useful for the community and the highest teacher absenteeism rate among surveyed systems. Beliefs about students matter equally. Across education systems, 23 percent of public officials believe that students' intelligence cannot change in meaningful ways. Evidence suggests that education systems guided by fixed expectations about students' learning potential tend to produce achievement disparities roughly twice as wide as systems that emphasize the possibility of development (the growth mindset).⁶⁸

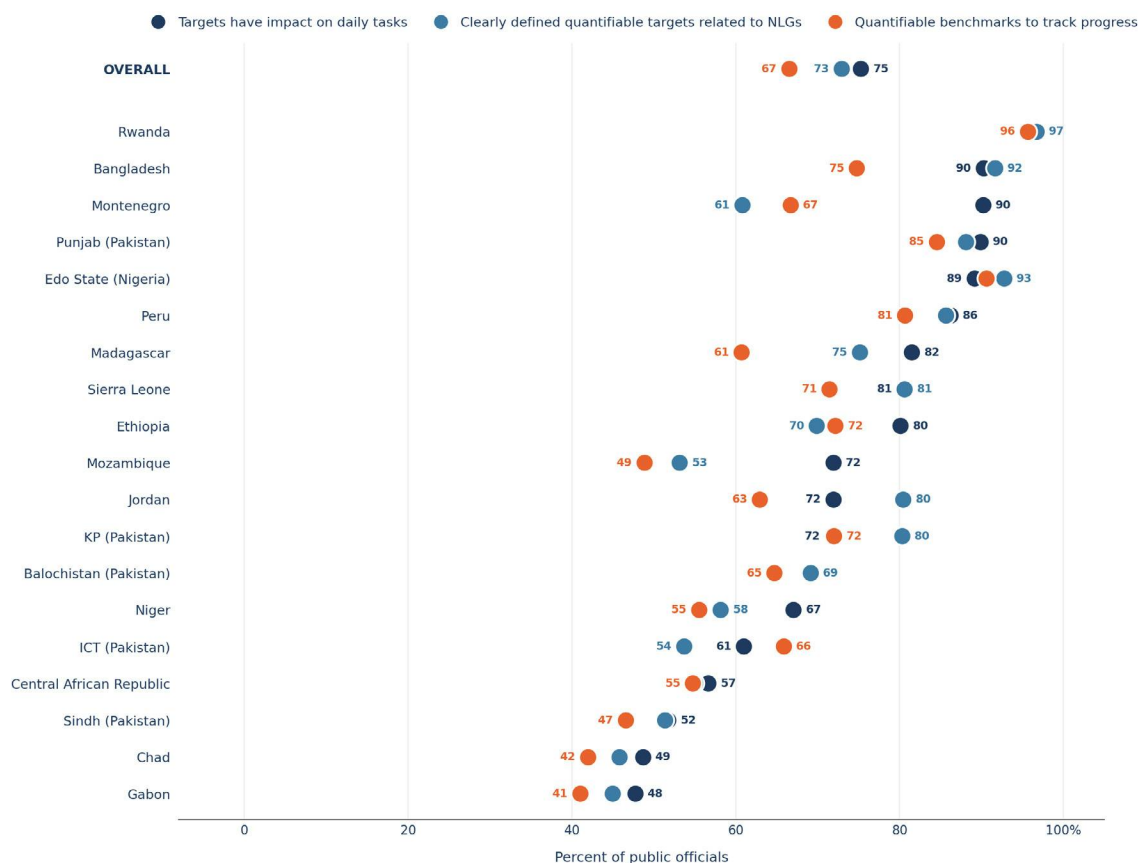
A disconnect persists between officials' perceptions and frontline realities. While officials believe teacher absenteeism is 11 percent, it stands at 22 percent.

Flying Blind: Officials Often Lack the Data Needed to Track and Act on Learning Goals

Most education bureaucracies have learning targets. These targets can help align systems with what matters most for improving learning outcomes. From the global Sustainable Development Goal 4.1 to country and subnational strategies, education systems increasingly articulate explicit learning ambitions.

The data reflects this commitment. Seventy-three percent of public officials surveyed report their organizations have clearly defined, quantifiable targets related to national learning goals (Figure 3.8). More importantly, these targets help shape daily work. Most officials report that their daily tasks are derived from organizational targets, meaning they understand what their organization aims to achieve and how their work contributes to achieving these targets. Most describe quantitative benchmarks that allow them to measure progress, though some acknowledge that these benchmarks remain limited or could be more detailed. The presence of learning targets across systems represents real progress. The language of learning has permeated administrative layers that once focused exclusively on inputs like enrollment and infrastructure. But targets alone are not enough. Their value depends on the monitoring systems to track progress, the incentives to reward achievement, and resource allocation decisions that align with learning priorities.

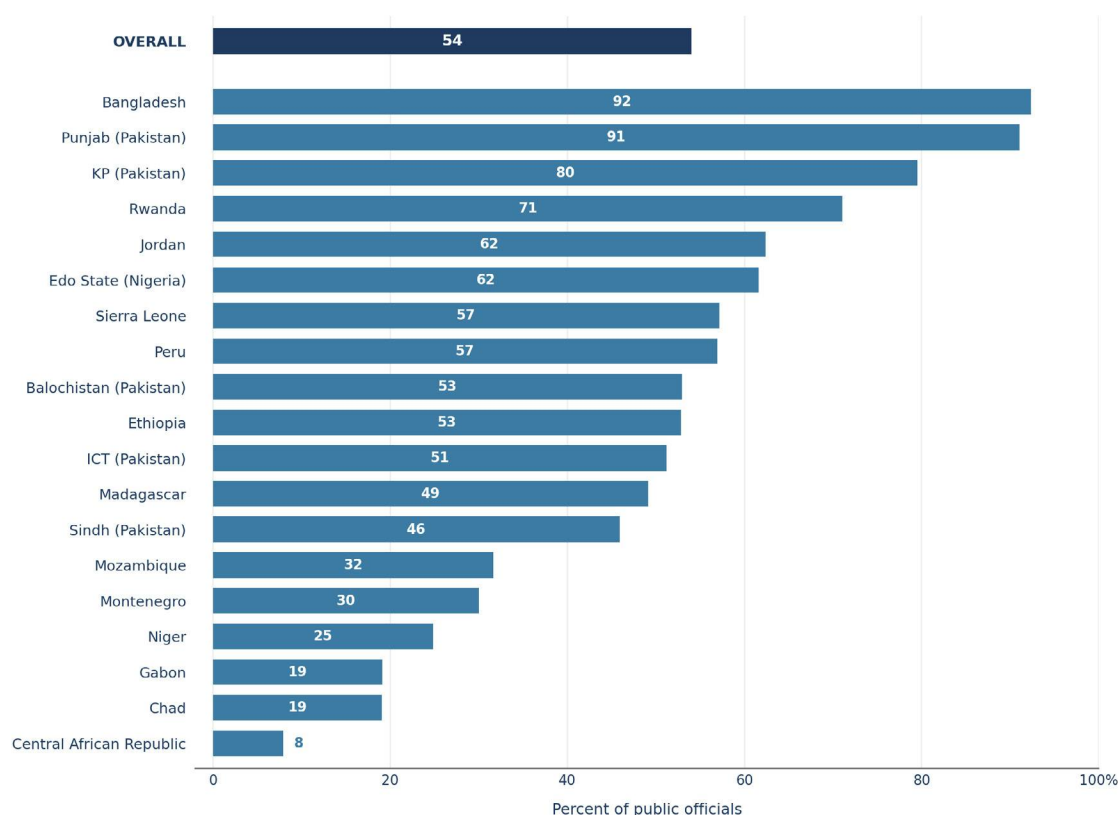
Figure 3.8: Existence and characteristics of organizational targets



Notes: "Overall" represents a simple average across education systems. Based on responses from 3,202, 3,203, and 3,201 officials regarding the alignment of organizational targets with the National Learning Goals (NLGs), measurability of organizational targets, and the effect of the targets on daily tasks respectively. 364, 363, and 365 officials are excluded due to missing or refused responses for each category.

Monitoring systems that can track progress toward learning targets function unevenly. Across the 19 education systems, 46 percent of public officials report lacking a functioning Management Information System (MIS), with the largest shares in the Central African Republic, Chad, and Gabon, where the overwhelming majority of officials report the absence of a functional MIS (Figure 3.9). Even when MIS systems do exist, the quality varies. Some respondents report reliable, recently updated systems with broad coverage across their areas of work, while others describe systems that function poorly or provide only limited coverage. Beyond data infrastructure, the frequency and scope of performance reviews also varies substantially. Some organizations conduct performance reviews several times per year with broad stakeholder participation. Others conduct limited annual reviews, or none at all.

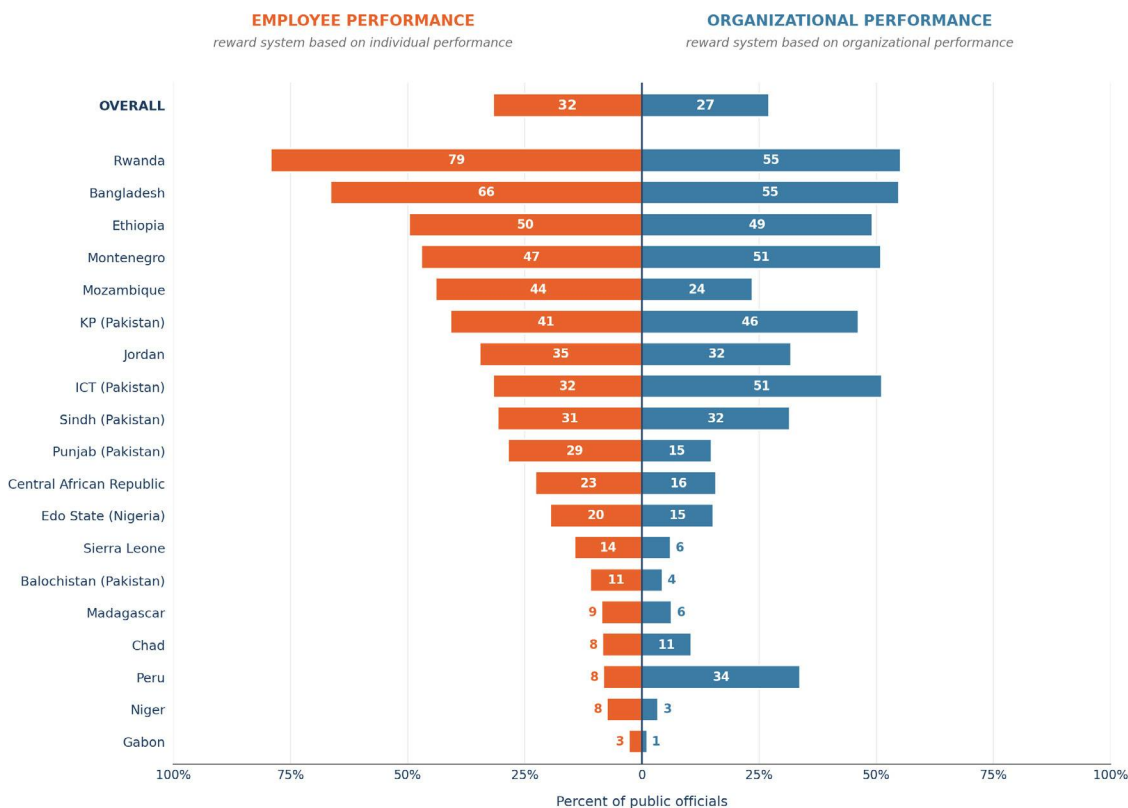
Figure 3.9: Availability of functioning management information systems



Notes: "Overall" represents a simple average across education systems. Based on responses from 3,198 officials; 368 officials are excluded due to no response.

The absence of robust monitoring weakens incentives to prioritize learning. Approximately 30 percent of officials report that formal reward systems exist for individual or organizational contributions toward national learning goals (Figure 3.10). Many describe contexts where individuals and units are treated identically regardless of performance. Some report that informal recognition may occur but without systematic evaluation criteria. Only in exceptional cases do officials describe formal evaluation systems where rewards (financial or nonfinancial) follow from well-defined and monitored achievements aligned to national learning goals, applied consistently across all staff and units. In Gabon, only one percent of officials report comprehensive incentive systems. In Rwanda, 55 percent do so. The practical consequences

Figure 3.10: Incentives to pursue national learning targets

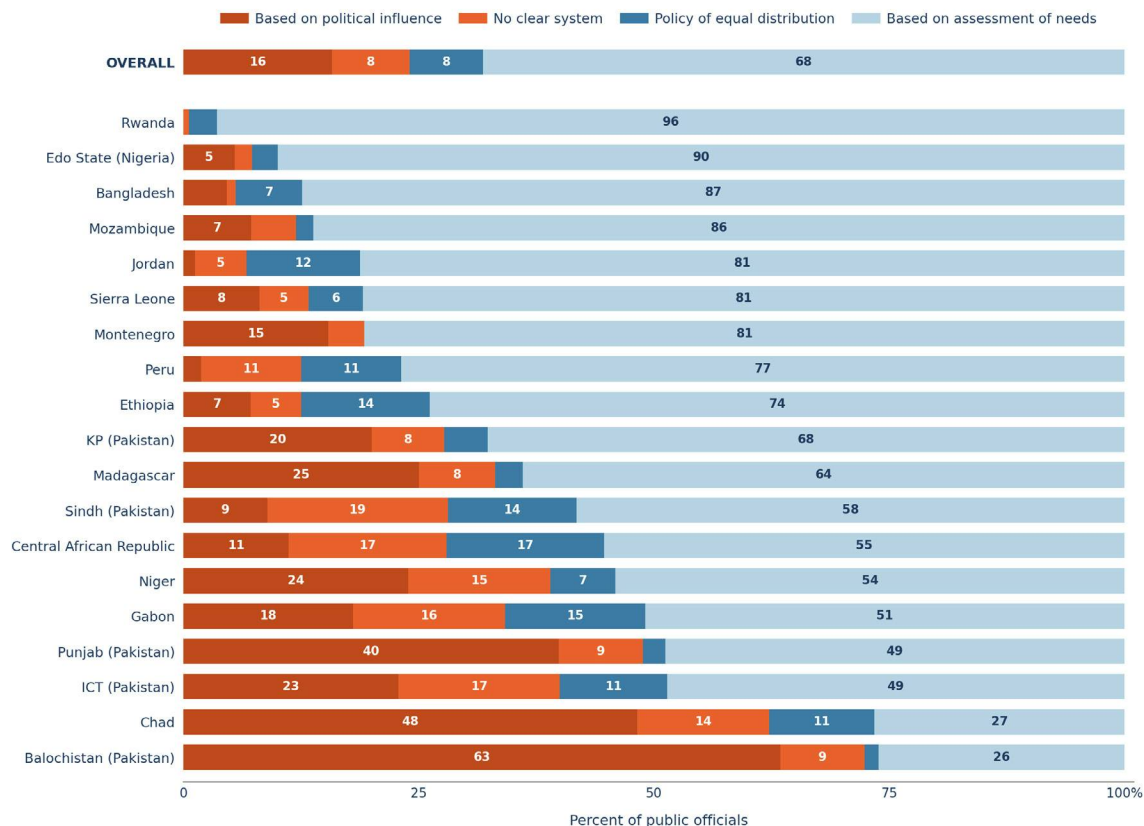


Notes: “Overall” represents a simple average across education systems. Each bar shows the share of public officials saying that there exists a formal and well-defined reward system for employee and organizational performance. Based on responses from 3,200 and 3,201 officials on employee and organizational performance respectively (366 and 365 officials respectively are excluded due to no response).

are straightforward. An education officer who successfully raises literacy rates receives the same recognition (or lack thereof) as one who does not. Under those conditions, learning targets lose their potential to drive improvement.

Weak incentives influence how systems allocate resources. The 2023 Education Finance Watch documented how low- and lower-middle-income countries often struggle to allocate educational funds efficiently, undermining the impact of their spending.⁶⁹ Officials across surveyed systems frequently describe processes that are disconnected from learning outcomes (Figure 3.11). In many systems, budgets are distributed evenly among schools or districts regardless of local conditions, or allocated through ad hoc decisions. Some report that budgets loosely reflect demographic indicators like enrollment, but these considerations rarely complement performance incentives. Budget allocations that systematically integrate indicators of need with incentives for performance remain the exception. In Rwanda, 96 percent of officials report that budget decisions consider both need and performance. In Chad, only 27 percent do. Without considering these critical factors, schools may receive per-student funding that does not reflect their differing needs.

Figure 3.11: Budget allocation decisions



Notes: "Overall" represents a simple average across education systems. Based on responses from 2,977 officials; 589 are excluded due to no response.

These weaknesses in the bureaucratic system show up in the quality of education that children receive. Bureaucracies that cannot reward learning improvements and ignore outcomes when allocating resources produce low learning quality across the education system. Education systems where bureaucratic quality is stronger tend to deliver better education services at the school level. For example, in Edo State, Nigeria, concerted efforts to strengthen bureaucratic systems and focus on learning outcomes have coincided with measurable improvements in classroom practices and student learning. In systems where bureaucratic dysfunction persists, schools struggle regardless of teacher effort or student potential. The quality of bureaucratic machinery does not guarantee learning, but its absence almost certainly prevents it.

4

The Path Forward



Countries that have successfully improved learning outcomes have often taken a systems approach to education reform.⁷⁰ Rather than implementing isolated interventions, these countries have pursued incremental, coordinated change across practices, policies, and politics, recognizing that alignment between these dimensions is critical for sustainable improvement. Countries that have successfully advanced learning outcomes have established clear learning goals, created robust monitoring systems to track progress, strengthened bureaucratic capacity for effective implementation, and developed mechanisms to ensure accountability at all levels. They have also sustained a consistent commitment to learning improvements over multiple political cycles, recognizing that sustainable educational change requires persistence beyond short-term political horizons. What distinguishes these success stories is not necessarily higher spending or more ambitious individual programs, but rather stronger coherence across different parts of the system, such that efforts at the classroom, school, district, and ministry levels reinforce rather than undermine each other.

Peru presents one such example. Ranking last among participating countries in the 2012 PISA assessment sparked comprehensive reforms across the education system, including curriculum revisions, teacher evaluation systems, and infrastructure investments.⁷¹ Among these reforms, the introduction of Semáforo Escuela in 2015 created a monitoring system that collects real-time data on teacher and student attendance, learning materials, and infrastructure, with information flowing to district, regional, and national officials to enable targeted support and policy adjustments.⁷² Within three years, teacher attendance improved significantly and learning outcomes began to rise. The experience of the Semáforo in Peru inspired the creation of the GEPD to ignite similar progress in other countries.

Yet Peru's success with Semáforo Escuela, and indeed any attempt at systems reform, rests on a foundation that too many countries lack: system-level visibility into how their education systems function. The Semáforo worked because it made the invisible visible, tracking key indicators in real time and feeding that information back to decision-makers at every level. But most countries cannot see what happens inside their schools, let alone understand how policies shape classroom practice or whether the bureaucratic capacity exists to implement reforms effectively. Policy makers know their education budgets and can measure student learning outcomes, but the crucial mechanisms connecting inputs to results remain largely unknown.

This limited visibility helps explain why so many interventions fail at scale. Reforms are often designed without a clear understanding of binding constraints, implemented without tracking classroom reach or fidelity, and sustained without feedback loops for course correction. The result is a policy environment where education spending does not translate into learning gains because system bottlenecks cannot be identified and addressed in a timely manner.

The GEPD provides comprehensive visibility into primary education systems, but the learning crisis does not end at primary school. As countries seek to build the skilled workforces their economies need, understanding how secondary, TVET, and higher education systems develop job-relevant skills becomes equally critical. The urgency extends beyond economics. An ambitious target, such as connecting 300 million people to electricity through Mission 300, or reaching 1.5 billion people with better health care, is unlikely to be achieved without workers who have the right skills to deliver it. Countries already draw on a range of tools to monitor performance across these subsectors, from EMIS and learning assessments to skills-specific instruments like the STEP survey and IFC's Vitae tool. What is currently missing is an integrated diagnostic that brings together service delivery, policy coherence, and political and bureaucratic dynamics for education beyond the primary level, in the way the GEPD does for foundational learning.

The GEPD offers a strong foundation to build on. Its Survey of Public Officials and Policy Survey can be adapted and expanded to cover higher levels of education, complementing rather than duplicating existing instruments. Do secondary systems support the development of advanced cognitive and technical skills? Do TVET programs align with employer demand? Are education systems preparing young people for the transition from school to work? Extending the GEPD's integrated approach to these questions would help governments diagnose constraints and prioritize reforms across the full skills development continuum.

Lowering data collection costs is both possible and necessary. The GEPD has already reduced costs by more than half compared to predecessor instruments through streamlined data collection. Yet comprehensive diagnostics remain out of reach for many countries. Further cost reductions are achievable through continued innovation in technology and survey design. Artificial intelligence offers promising pathways. Automated coding of open-ended responses and machine learning algorithms to analyze classroom observation videos could significantly reduce the time and expertise required for data processing. Digital data collection methods, from remote video submissions for classroom observations to interviews with teachers and principals through secure platforms, can lower fieldwork costs while maintaining quality. Most importantly, there is scope to further streamline the core diagnostic itself by developing a lean, modular framework built around essential policy and governance indicators that countries can implement affordably and on a regular basis, with optional modules for those seeking deeper analysis.

5

Conclusion



Policy makers in low- and middle-income countries who are working to improve student learning outcomes often find themselves operating without critical data to guide decisions. They see the budgets allocated to education and, in some cases, the learning that students acquire. But they lack information on many of the crucial factors in between, including the practices, policies, and politics that drive learning outcomes. This limited visibility has allowed the learning crisis to persist even as school enrollment has expanded. Millions of children attend classrooms for years without acquiring basic skills, and systems lack the data to understand what holds them back. The preceding chapters have provided a cross-country analysis of key drivers of learning, spanning practices, policies, and political economy factors, drawing on data from School Survey data from 18 education systems and Public Officials' data from 19 education systems. The analysis reveals not just isolated problems but the interconnected factors that determine whether children learn.

The GEPD was designed to provide countries with comprehensive system-level diagnostics across practices, policies, and politics. Since its development, the GEPD has been implemented in 27 education systems, representing around 18 percent of World Bank IDA/Blend countries. The data have proven valuable not just for diagnosis but for action, informing country partnership frameworks, public expenditure reviews, GPE country compacts, and project design. Both Ethiopia and Peru have implemented second rounds to measure changes over time. In Sierra Leone, data on teacher pedagogical practices informed the design of continuous professional development interventions, with classroom videos from the TEACH observation tool used to facilitate “teacher circles” for peer learning. In Edo State, Nigeria, the data validated the strengths of the government’s EdoBEST program while identifying gaps that prompted a stronger focus on early childhood education. In Bangladesh, GEPD findings shaped the design of the government’s Primary Education Development Program, with interventions emphasizing expanded school readiness and strengthened teacher support. In Pakistan’s Balochistan province, the results sparked discussions on reforms and helped spotlight poor foundational learning outcomes, generating government interest in strengthening early childhood development and teacher professional development.

The GEPD’s cross-country comparability offers an additional advantage. Countries can benchmark their performance against peers and learn from systems facing similar challenges. Comparisons showing that teachers in similar contexts achieve higher pedagogical scores, or that systems at similar income levels provide better infrastructure, challenge assumptions about what is possible and create pressure for improvement. This comparative dimension transforms the GEPD from a purely diagnostic tool into a catalyst for reform, helping countries identify not just their weaknesses but practical examples of how other systems have addressed them.

Countries have succeeded in getting children through school doors at unprecedented rates, but those same systems often fail to ensure that time spent in classrooms translates into learning. The Learning Poverty statistics highlight this failure. Simulated estimates show that 70 percent of children in low- and middle-income countries are unable to read and understand simple text by age 10. The GEPD data show why low learning outcomes persist across diverse contexts.

At the school level, challenges start with limited instructional time. Across most systems, more than 15 percent of teachers are absent from the classroom during scheduled teaching time. Similarly, across most systems, at least 15 percent of students are absent during unannounced school visits. But even when teachers and students show up, the data reveal key challenges constraining the acquisition of foundational skills.

- **Many students arrive at school unprepared to thrive in primary education.** Only in three systems (Bangladesh, Montenegro, and Peru) do more than 50 percent of the students demonstrate mastery of basic skills needed to succeed in primary school. Student preparedness varies considerably across

contexts, from single digits in some systems to more than 70 percent in Montenegro. Numeracy skills tend to be relatively stronger, while socio-emotional skills often lag behind.

- **Teachers often lack essential content knowledge and pedagogical skills.** Overall, across most systems surveyed, less than 20 percent of teachers demonstrate proficiency in the subjects they teach. A majority of teachers across most education systems do not have satisfactory pedagogical practices, with considerable variation across surveyed systems. While some key aspects of support, such as classroom observations, are relatively common, allocating more time to post-observation feedback conversations and increasing the use of lesson plans would strengthen pedagogical practice.
- **Progress on providing basic inputs and infrastructure remains uneven.** Basic classroom essentials like blackboards and desks are largely available. But access to essential learning materials remains limited. Across a majority of the systems surveyed, less than 80 percent of Grade 4 students have textbooks, as observed by an enumerator, with significant variation across contexts. Access to electricity, water, and functional toilets varies widely across systems, with deficits disproportionately affecting rural areas.
- **Principals often lack an accurate picture of their own schools.** School leaders systematically overestimate teachers' content knowledge, making it difficult to identify and address teaching gaps. Access to mentoring or coaching from experienced principals remains scarce. No surveyed education system offers such opportunities to more than half of its principals.

At the policy level, the GEPD shows that many of the challenges observed at the school level stem from underlying policy-level weaknesses.

- **Implementation gaps are pervasive.** While most surveyed systems have comprehensive policy frameworks across critical dimensions, many policies remain weakly implemented in practice. Across most systems, a practicum is required as part of preservice training, yet only 53 percent of teachers in those systems report completing one. Most systems have policies for in-service training,⁷³ yet participation rates range from four percent in Pakistan's Balochistan province to 72 percent in Peru.
- **Key policy frameworks are missing entirely in critical areas.** Only four systems have policies governing education technology. As digital technology becomes an increasingly important part of the teaching and learning process, establishing enabling policies and guiding frameworks will be essential.
- **The political and bureaucratic context often explains why well-designed policies fail to translate into improved practice.** The GEPD data challenge a common narrative, and highlight that the problem is not that education officials lack capacity or commitment. Officials are highly educated (59 percent hold a master's degree or higher), and 81 percent report satisfaction working in government, with only 27 percent wanting to switch to the private sector in the next two years. The issue is that even capable, dedicated officials struggle to improve learning outcomes when the systems around them are misaligned with learning goals.
- **Instability undermines sustained reform.** The education systems surveyed have experienced high ministerial turnover over the past six years, with Peru cycling through 12 ministers in that period. More than half of all department heads changed within the 12 months preceding the GEPD implementation. This frequent turnover means institutional memory erodes rapidly, and long-term reforms lose momentum before they can take root.
- **Political interference distorts personnel decisions.** One-quarter of officials report that hiring decisions favor political power over merit, and 23 percent report the same for promotions. Ignoring merit in staffing weakens education systems' ability to recognize and reward skilled personnel. Around 30 percent of all public officials believe that serious misconduct, from distorting procurement to hiring based on favoritism to reporting false information, would face no formal consequences. Without meaningful accountability, incentives to perform well erode.

- **Learning is not measured or rewarded.** Approximately half of officials report that they lack a functioning management information system. Fewer than half say their organizations reward contributions toward national learning goals. Systems that cannot track learning outcomes or reward improvement leave officials without the information or incentives to prioritize what matters for uplifting education systems.
- **There is a striking disconnect between officials' perceptions and frontline realities.** Officials overestimate class sizes by 83 percent and underestimate teacher absenteeism by half. Officials cannot solve problems they do not accurately perceive.

While decades of rigorous evidence demonstrate what works to improve foundational learning, the exact solutions will need to be tailored to country context. The GEPD data clearly illustrate why. For example, in Chad and Sierra Leone, most of the variation in learning outcomes is between schools: where a child is enrolled largely determines what they learn. In other contexts, such as Jordan and Pakistan's ICT, schools are more similar to each other on average, but classrooms contain students at very different learning levels. These are not just diagnostic distinctions. They point to the need to contextualize and adapt reform priorities: uplifting entire schools by tackling constraints that keep most students in the class from learning (for example, overambitious curricula or inadequate teacher support) versus investing in targeted instruction and differentiated support where in-class variation in learning outcomes is high. What is common across all systems is that without data to reveal where the problem lies, even well-re-sourced reform efforts risk missing the mark.

Addressing the challenges constraining foundational learning require system-level data. Education systems often have the committed workforce, the policy frameworks, and in many cases the financial resources to improve learning. What they often lack is system-level visibility, leaving binding constraints unidentified and scale-up efforts fragile. Although every context needs interventions tailored to its unique local challenges, data such as those provided by the GEPD paints a full picture of the system and can help identify learning bottlenecks relatively quickly and cost effectively. With timely evidence guiding decisions, systems can lift foundational skills at scale, and bring Fatima, and millions like her, within reach of the futures they aspire to.



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